AMSE forecast quality across weather features

Original GC · MSE · AMSE λ=0.1 · Full AMSE

This report compares spectral power, forecast–truth coherence and deterministic forecast error using completed evaluations of the same 1° residual-Mamba architecture. The three trained models use stochastic weight averaging over updates 500, 1000, 1500 and 2000.

Full AMSE restores substantially more missing spectral power, but it also loses pattern alignment and much of the MSE-trained model's deterministic accuracy advantage. AMSE λ=0.1 stays much closer to MSE: its spectral changes are smaller and its forecast-loss performance is similar.

Model	Mean relative power error	Mean spectral coherence
Original GC	48.94%	0.5115
MSE	57.16%	0.5244
AMSE λ=0.1	55.58%	0.5242
Full AMSE	31.96%	0.4902

These spectral summaries give equal weight to 11 fields, zonal wavenumbers 1–180 and days 1, 3, 7 and 10. Raw spectra are pooled across eight 2023 starts before ratios are formed. The summaries include wind speed alongside its components and are secondary diagnostics, not the original GraphCast training loss.

The report contains a methods guide, spectral summary tables, exact GraphCast loss, global RMSE by variable, and three spectral plots for each field. The downloadable tables include every saved lead and all 83 global-RMSE channels.

Evidence scope: 32 historical 2022 development starts and eight exploratory 2023 starts, reported separately. No uncertainty intervals or new forecasts are included.

How to read the comparison

Definitions and fixed model references

Original GC is the baseline evaluation paired with the MSE SWA checkpoint from the original full-AMSE campaign. MSE is that same campaign's trained control. AMSE λ=0.1 comes from the later λ=0.1 campaign; Full AMSE comes from the original campaign. These four references are fixed throughout. Repeated GC evaluations are not averaged or silently substituted.

The separate λ=0.1 campaign has another matched MSE control. It is excluded from the four-model figures as requested, but retained in the source audit. Small cross-campaign differences may reflect run-to-run variability; the repeatability appendix records those differences.

Spectral diagnostics use zonal Fourier transforms of scalar fields within 30–60° latitude in both hemispheres. The zonal mean is removed at each latitude. Latitude weights are cosine weights, and raw powers and cospectra are averaged over starts before forming ratios. Curves are unsmoothed. Larger zonal wavenumber means finer scale, but wavelength depends on latitude.

Power error (%) = 100 × |forecast power − truth power| / truth power; lower is better. Coherence = real cospectrum / sqrt(forecast power × truth power); higher is better. ERA5 is a verification reference, not an additional forecast model.

Global RMSE is sqrt(mean squared physical error over starts and the global cosine-weighted grid). Upper-air variable-group RMSE additionally pools squared errors equally across 13 levels. Specific-level RMSE uses the stated pressure level. Neither metric is identical to pressure-weighted, difference-standard-deviation-normalized original GraphCast loss.

Exact GraphCast loss reduction is 100 × (1 − aggregate model loss / aggregate reference loss). Aggregation precedes the ratio. Displayed-reference and own-paired-reference comparisons are labeled separately. Wind-speed global RMSE was not saved; component RMSEs cannot reconstruct it. WS10 and WS850 have spectral diagnostics only.

Relative power error by field

Mean over days 1, 3, 7, 10 and k = 1–180 · percent · lower is better

Field	Original GC	MSE	AMSE λ=0.1	Full AMSE
T2m 2 m temperature	18.53	19.79	19.06	12.00
U10 10 m zonal wind	53.19	61.23	59.78	41.34
V10 10 m meridional wind	52.26	62.38	60.86	38.19
MSLP Mean sea-level pressure	17.26	22.76	21.08	14.70
TP6 6 h precipitation	67.69	81.09	78.22	26.40
T850 850 hPa temperature	47.39	52.64	51.02	25.83
Q700 700 hPa specific humidity	69.06	76.69	75.25	35.39
U850 850 hPa zonal wind	59.49	67.00	65.72	44.05
V850 850 hPa meridional wind	59.46	68.78	67.60	46.51
WS10 10 m wind speed	40.57	52.48	50.36	25.20
WS850 850 hPa wind speed	53.40	63.93	62.46	42.00

Each entry gives equal weight to the retained spectral bins, rather than weighting by their physical energy. Do not interpret these means as global RMSE or GraphCast loss. The individual field plots show scale and lead dependence that these averages conceal.

Spectral coherence by field

Mean over days 1, 3, 7, 10 and k = 1–180 · higher is better

Field	Original GC	MSE	AMSE λ=0.1	Full AMSE
T2m 2 m temperature	0.8606	0.8658	0.8652	0.8462
U10 10 m zonal wind	0.5044	0.5223	0.5217	0.4882
V10 10 m meridional wind	0.4901	0.5104	0.5092	0.4817
MSLP Mean sea-level pressure	0.7607	0.7660	0.7661	0.7226
TP6 6 h precipitation	0.2895	0.3016	0.3055	0.2683
T850 850 hPa temperature	0.5769	0.5876	0.5867	0.5533
Q700 700 hPa specific humidity	0.3161	0.3331	0.3322	0.2964
U850 850 hPa zonal wind	0.4336	0.4441	0.4437	0.4063
V850 850 hPa meridional wind	0.4287	0.4439	0.4426	0.4116
WS10 10 m wind speed	0.5281	0.5435	0.5429	0.5103
WS850 850 hPa wind speed	0.4382	0.4503	0.4499	0.4073

Each entry gives equal weight to the retained spectral bins, rather than weighting by their physical energy. Do not interpret these means as global RMSE or GraphCast loss. The individual field plots show scale and lead dependence that these averages conceal.

Rollout RMSE by variable in 2022

32 starts · all 40 leads pooled in squared-error space · lower is better

Variable group	Original GC	MSE	AMSE λ=0.1	Full AMSE
2 m temperature (K)	1.62	1.519	1.522	1.618
10 m zonal wind (m/s)	2.699	2.395	2.403	2.587
10 m meridional wind (m/s)	2.844	2.535	2.547	2.746
Sea-level pressure (hPa)	4.012	3.672	3.632	3.79
6 h precipitation (mm)	1.872	1.74	1.752	2.203
Temperature, all levels (K)	1.896	1.766	1.767	1.898
Specific humidity, all levels (g/kg)	0.7363	0.6784	0.6829	0.7712
Zonal wind, all levels (m/s)	5.312	4.868	4.886	5.294
Meridional wind, all levels (m/s)	5.52	5.03	5.05	5.457
Geopotential, all levels (m^2/s^2)	422.2	393.6	394.2	413
Vertical velocity, all levels (Pa/s)	0.127	0.1197	0.1205	0.1505

Largest Full AMSE rollout-RMSE increases versus MSE: 6 h precipitation +26.6%; vertical velocity, all levels +25.7%; specific humidity, all levels +13.7%.

These are physical-unit RMSEs, not the exact GraphCast loss. Upper-air groups pool all 13 levels equally. Small AMSE λ=0.1 differences are inconclusive without uncertainty estimates.

Rollout RMSE by variable in 2023

8 starts · all 40 leads pooled in squared-error space · lower is better

Variable group	Original GC	MSE	AMSE λ=0.1	Full AMSE
2 m temperature (K)	1.679	1.574	1.577	1.707
10 m zonal wind (m/s)	2.75	2.505	2.518	2.711
10 m meridional wind (m/s)	2.854	2.596	2.607	2.818
Sea-level pressure (hPa)	3.928	3.672	3.628	3.858
6 h precipitation (mm)	2.17	2.073	2.078	2.468
Temperature, all levels (K)	1.877	1.78	1.776	1.906
Specific humidity, all levels (g/kg)	0.7695	0.716	0.7198	0.8054
Zonal wind, all levels (m/s)	5.293	4.941	4.959	5.377
Meridional wind, all levels (m/s)	5.461	5.112	5.11	5.57
Geopotential, all levels (m^2/s^2)	422.4	394.8	394.1	421.9
Vertical velocity, all levels (Pa/s)	0.154	0.1492	0.1496	0.1748

Largest Full AMSE rollout-RMSE increases versus MSE: 6 h precipitation +19.0%; vertical velocity, all levels +17.1%; specific humidity, all levels +12.5%.

These are physical-unit RMSEs, not the exact GraphCast loss. Upper-air groups pool all 13 levels equally. Small AMSE λ=0.1 differences are inconclusive without uncertainty estimates.

Why sharper power can coexist with larger error

Measured mechanism and comparison with the AMSE paper

Power matching does not ensure that a feature is correctly located or timed. In each spectral mode, ordinary squared error equals an amplitude-mismatch term plus a decorrelation term. Restoring the amplitude of uncertain patterns can increase error even if coherence stays fixed.

$$\mathrm{MSE}_k = (\sqrt{P_f} - \sqrt{P_t})^2 + 2\sqrt{P_f P_t}(1 - \rho_k)$$

Day 10 T850	Amplitude error (K²)	Decorrelation (K²)	Total anomaly MSE (K²)
Original GC	0.301	17.203	17.504
MSE	1.336	12.388	13.724
AMSE λ=0.1	1.137	12.522	13.659
Full AMSE	0.480	15.704	16.184

These are midlatitude zonal-anomaly errors, not global T850 RMSE. The decorrelation term increases with both larger power and reduced coherence. AMSE includes a coherence penalty; finite network capacity, optimization and autoregressive dynamics still permit worse alignment on evaluation cases.

The original paper's main forecast-skill comparison uses nine-member lagged ensembles: CRPS improves while ensemble-mean RMSE changes little. This is a different target from individual-forecast RMSE. Its training also differs: 0.25° GraphCast, 32,500 updates, and a single-step-to-12-step curriculum. Here a frozen 1° GraphCast supports a residual branch trained for 2,000 updates with stopped physical-feedback gradients. These are plausible contributors, not isolated causal findings.

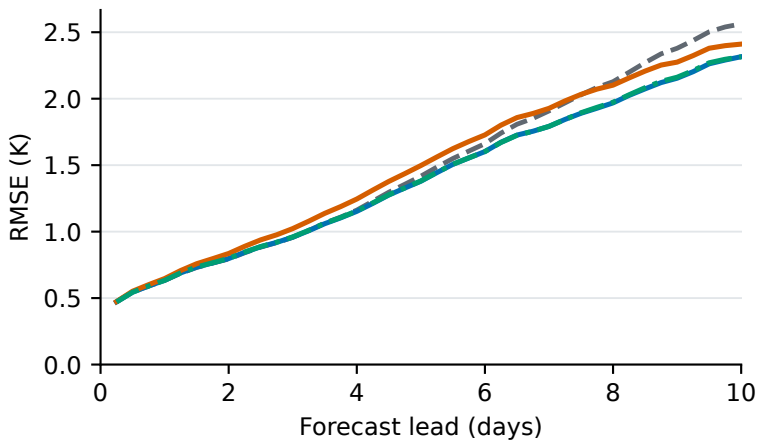
Source: Subich et al., sections 2.4 and 3.2 · arxiv.org/html/2501.19374v2

Forecast RMSE by variable

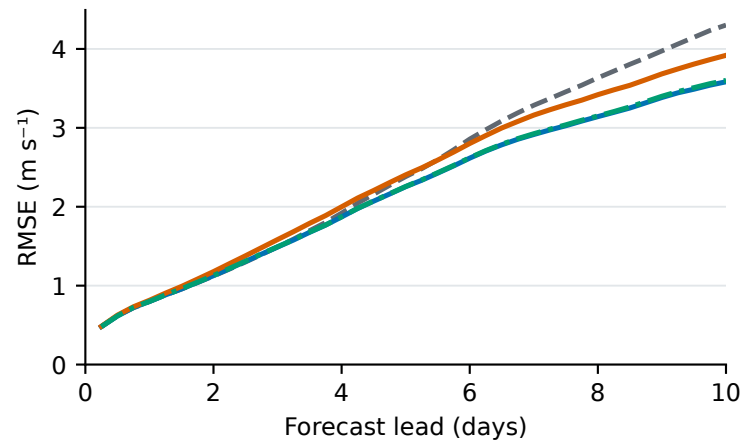
32 matched 2022 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

Original GC · MSE · AMSE λ=0.1 · Full AMSE

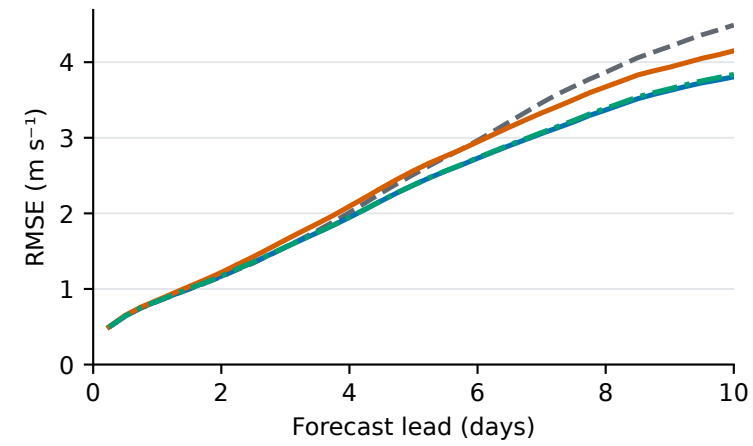
2 m temperature



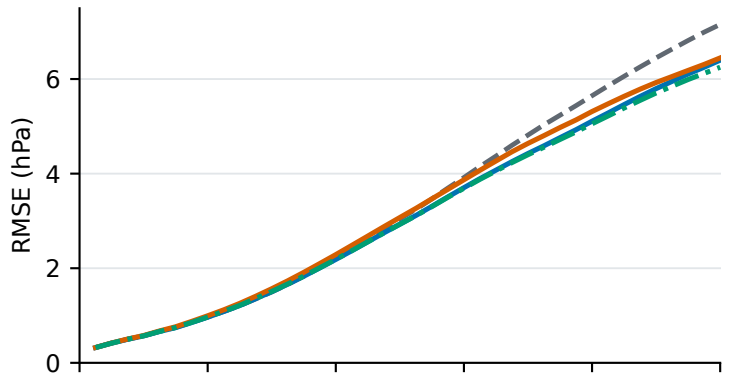
10 m zonal wind



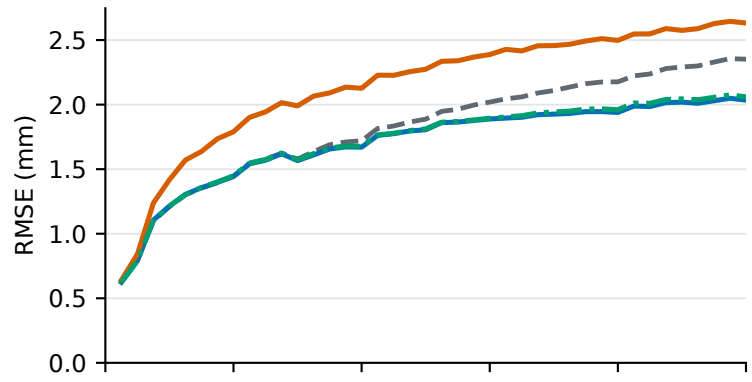
10 m meridional wind



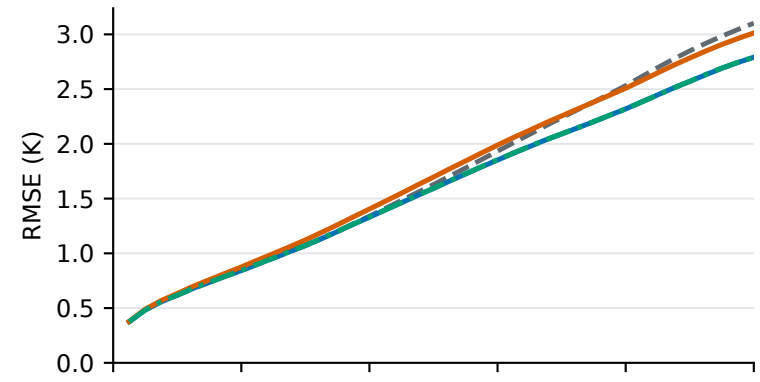
Mean sea-level pressure



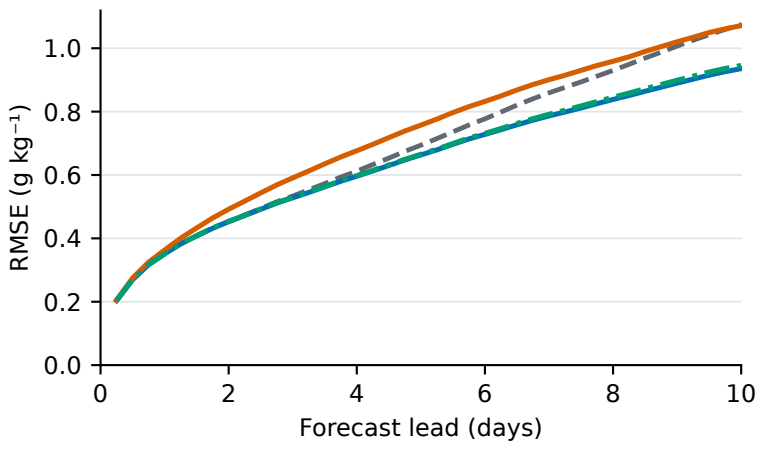
6 h precipitation



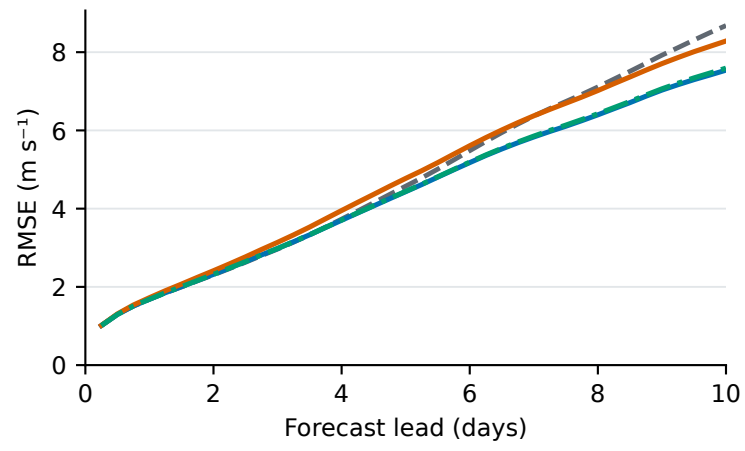
Temperature · all pressure levels



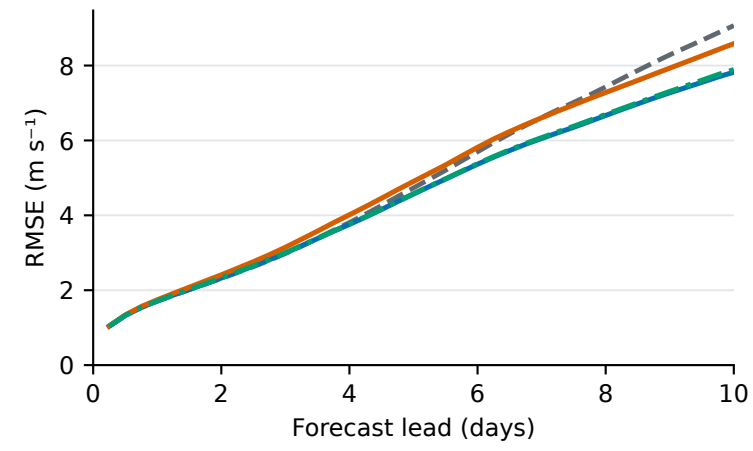
Specific humidity · all levels



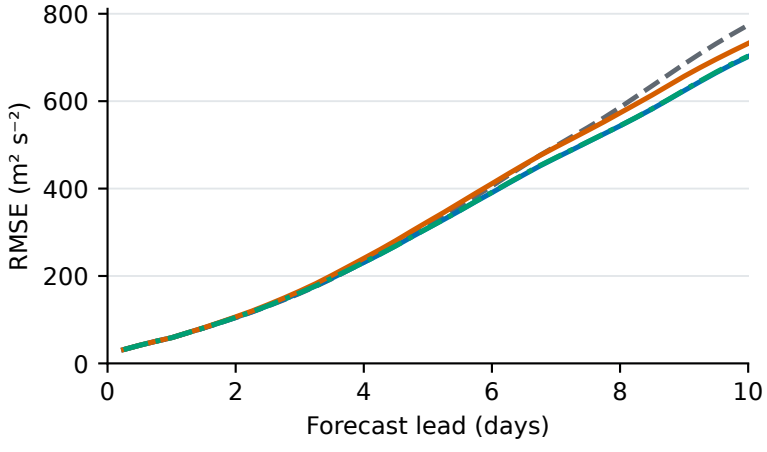
Zonal wind · all pressure levels



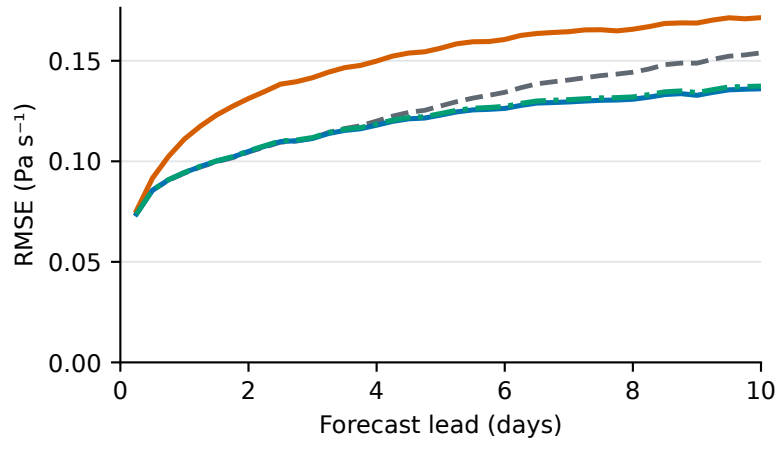
Meridional wind · all levels



Geopotential · all pressure levels



Vertical velocity · all levels



How to read these panels

Lower RMSE is better.
Each panel has its own physical units.

Upper-air groups pool squared error equally across all 13 pressure levels before taking the square root.

Global latitude-weighted errors pooled over the saved starts. These physical-variable RMSEs differ from the exact weighted GraphCast loss.

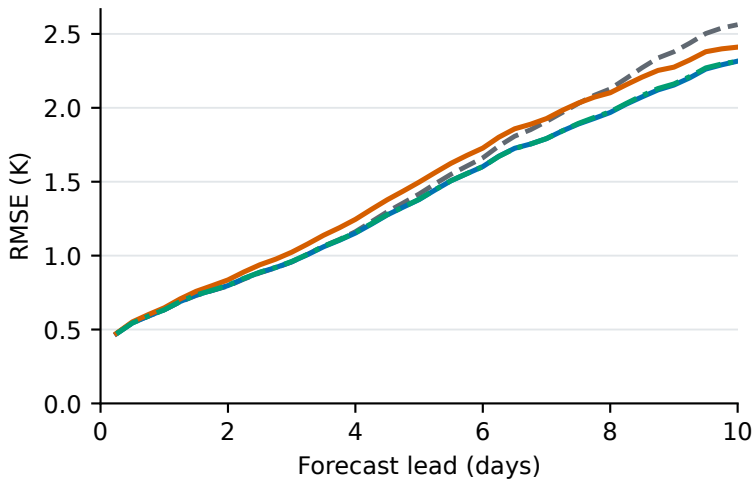
One common Original GC reference is shown; repeated baseline evaluations have small differences. No uncertainty intervals are available.

Global RMSE for the spectral diagnostic fields

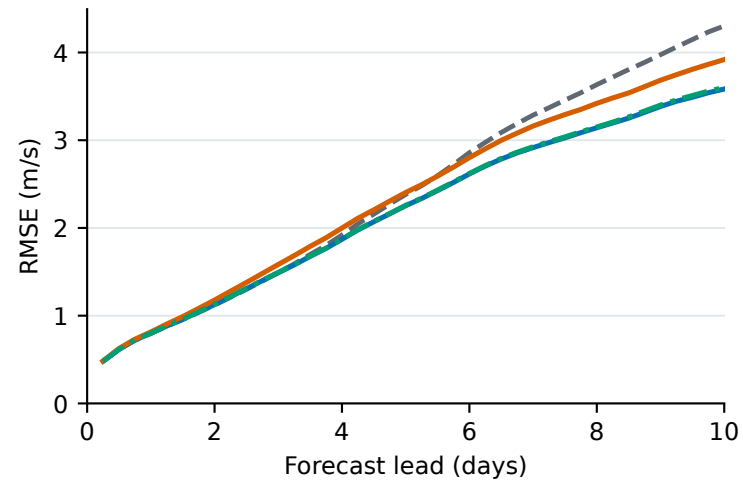
32 matched 2022 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

Original GC · MSE · AMSE λ=0.1 · Full AMSE

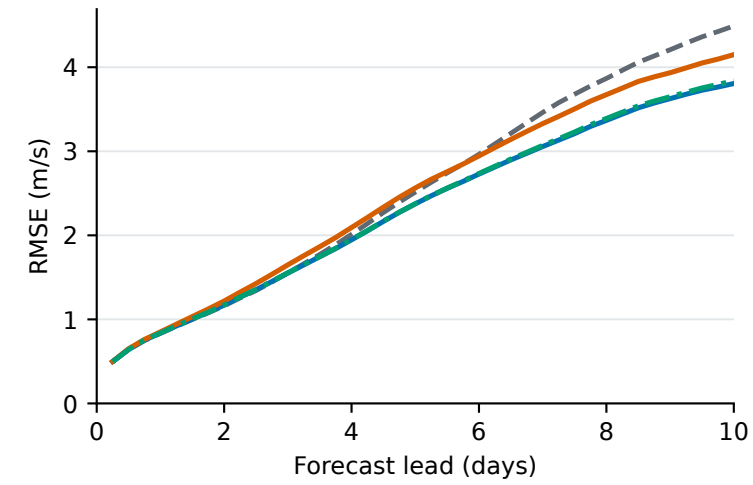
T2m · 2 m temperature



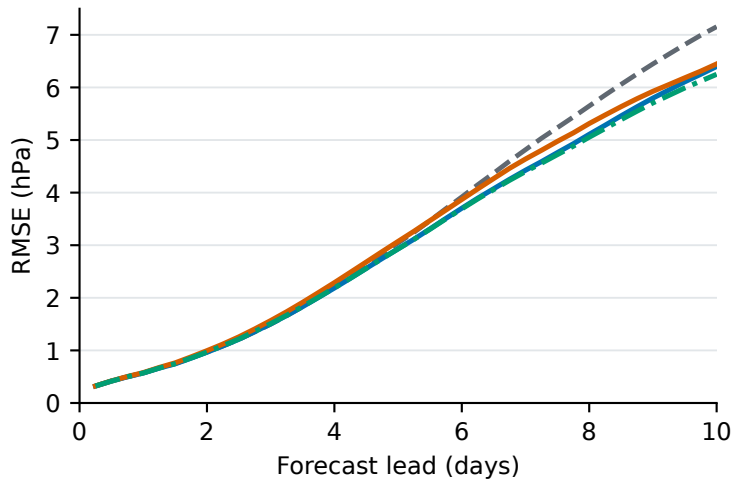
U10 · 10 m zonal wind



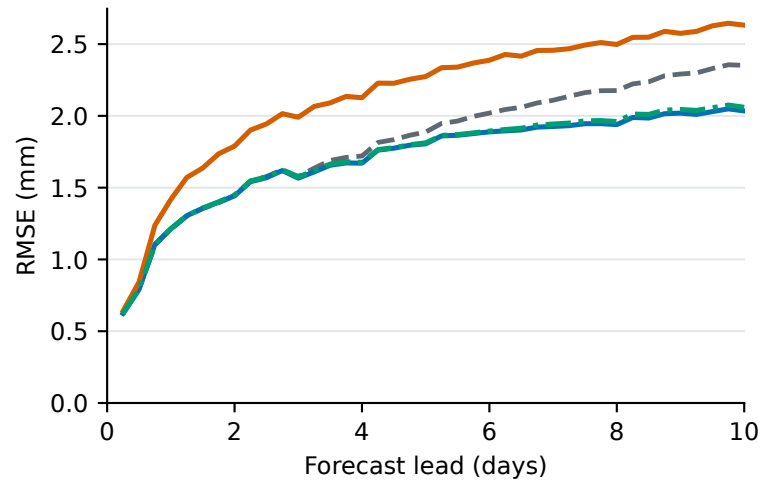
V10 · 10 m meridional wind



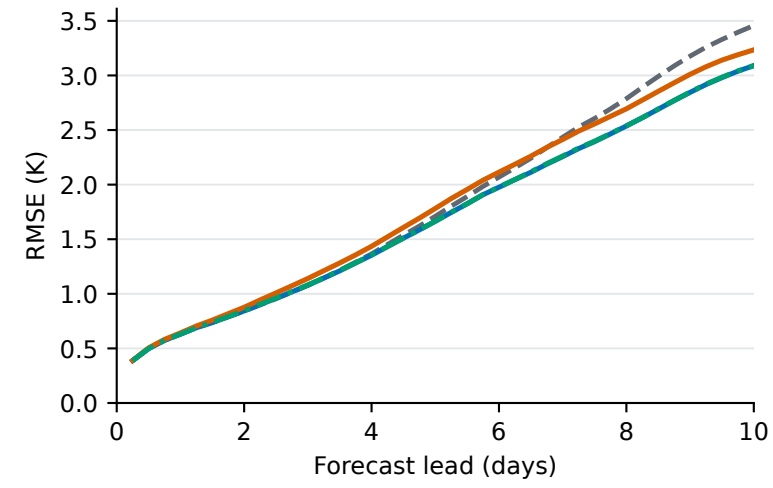
MSLP · Mean sea-level pressure



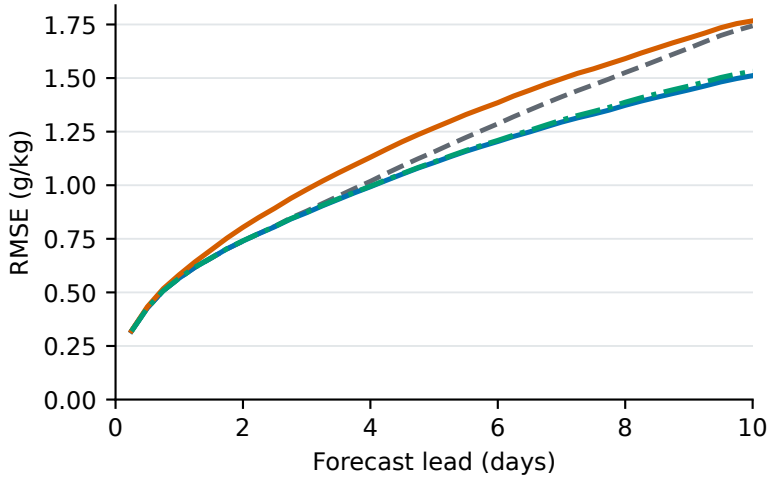
TP6 · 6 h precipitation



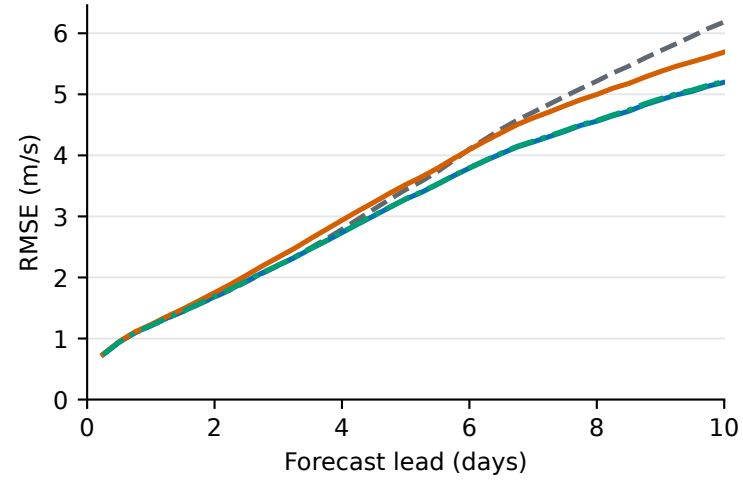
T850 · 850 hPa temperature



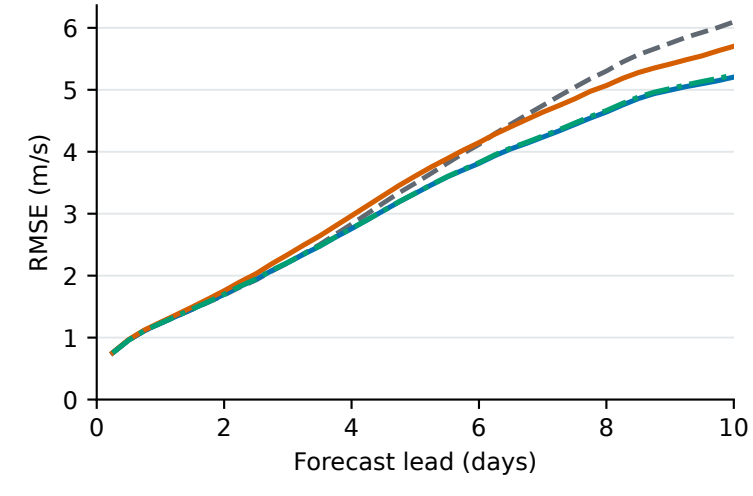
Q700 · 700 hPa specific humidity



U850 · 850 hPa zonal wind



V850 · 850 hPa meridional wind



Global latitude-weighted physical RMSE at the specified level; the companion spectral curves use only 30–60° latitude in each hemisphere.

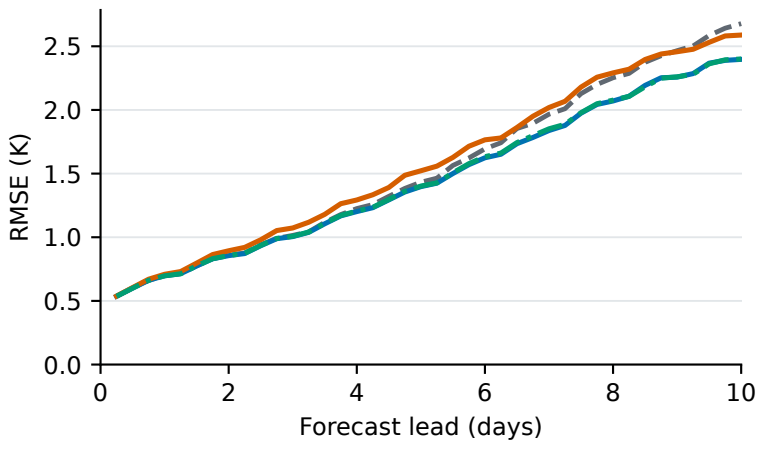
Global wind-speed RMSE (WS10 / WS850) was not saved; it cannot be reconstructed from component RMSEs. No uncertainty intervals.

Forecast RMSE by variable

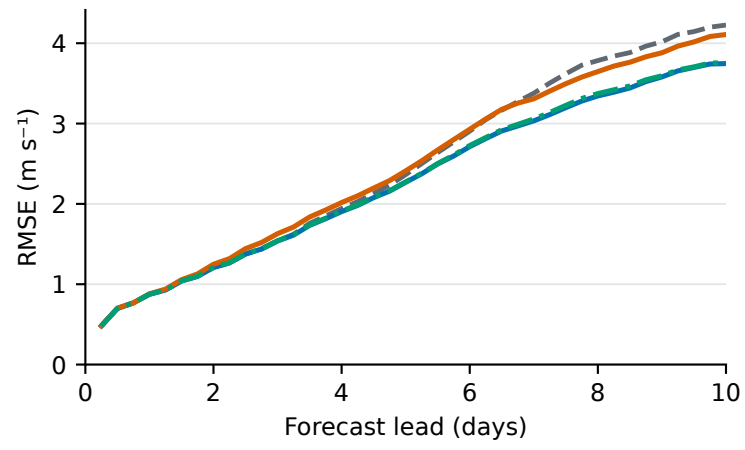
8 matched 2023 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

Original GC · MSE · AMSE λ=0.1 · Full AMSE

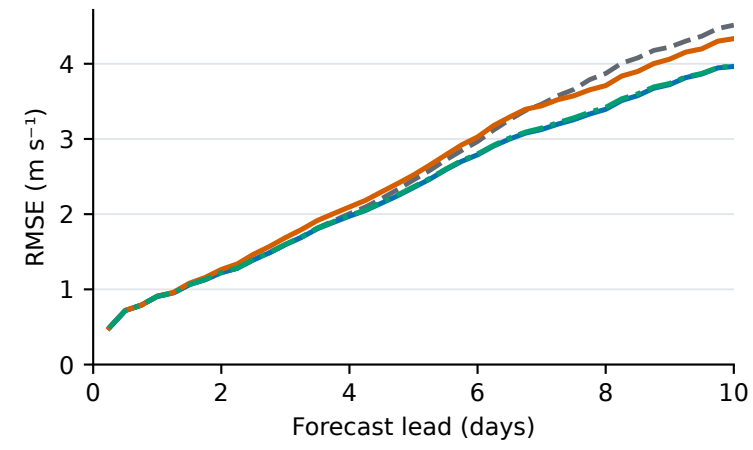
2 m temperature



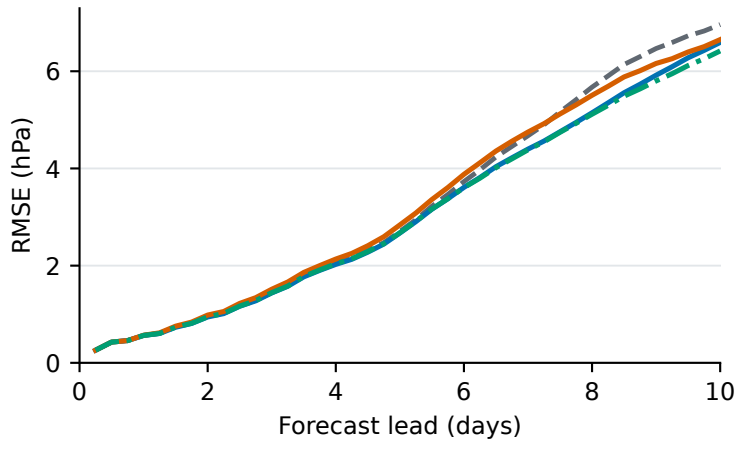
10 m zonal wind



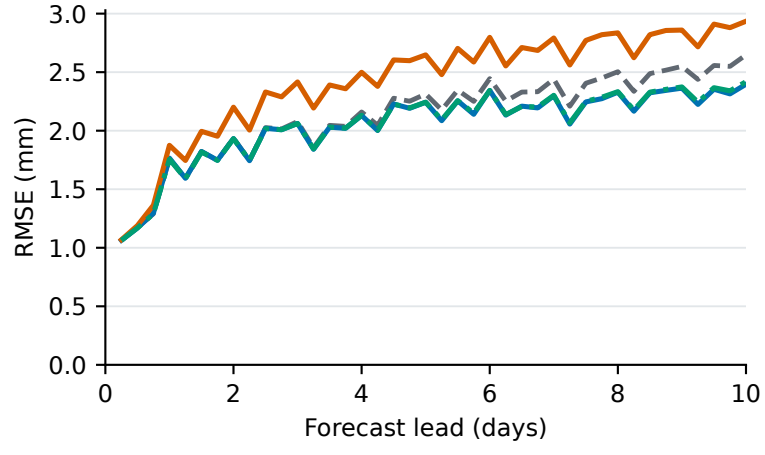
10 m meridional wind



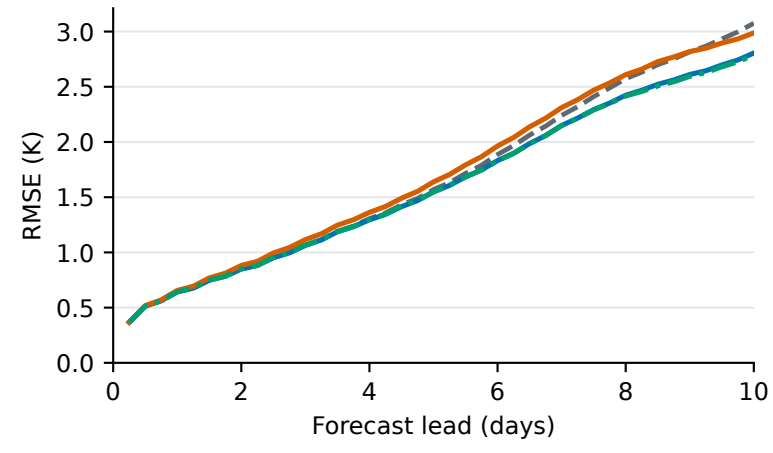
Mean sea-level pressure



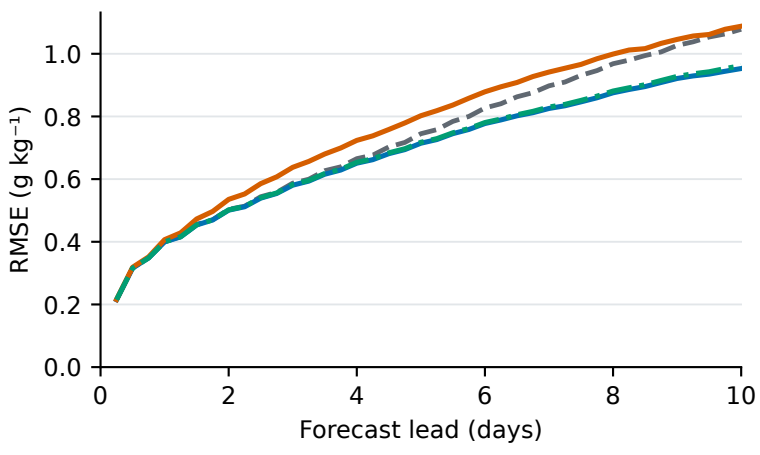
6 h precipitation



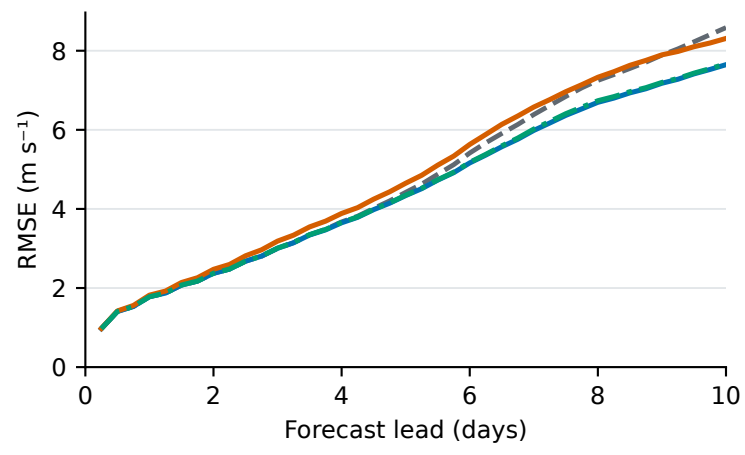
Temperature · all pressure levels



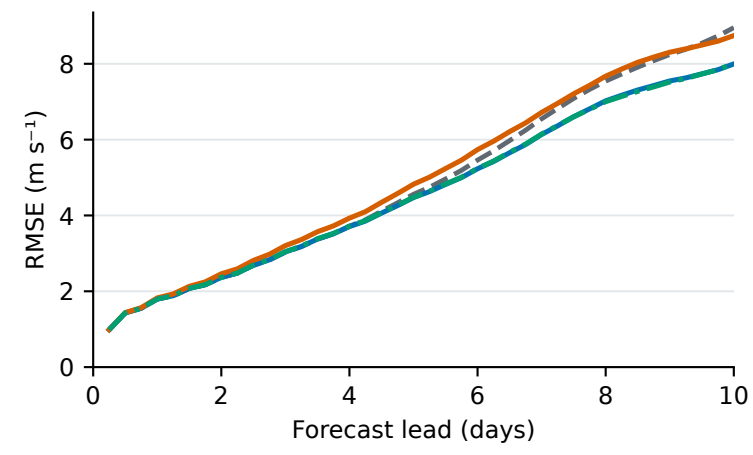
Specific humidity · all levels



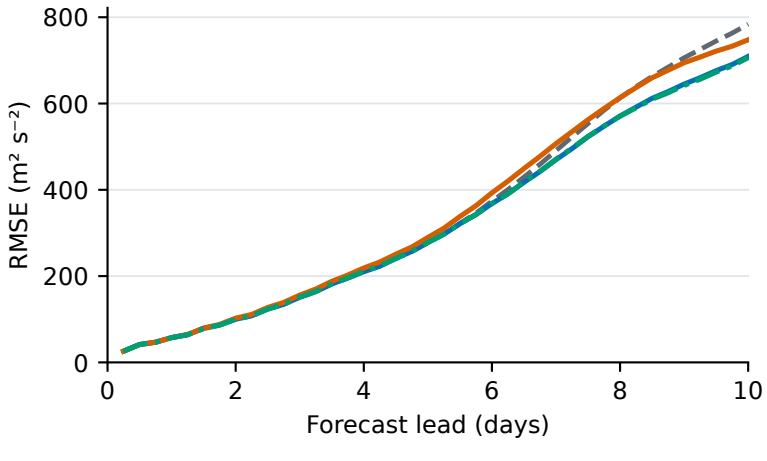
Zonal wind · all pressure levels



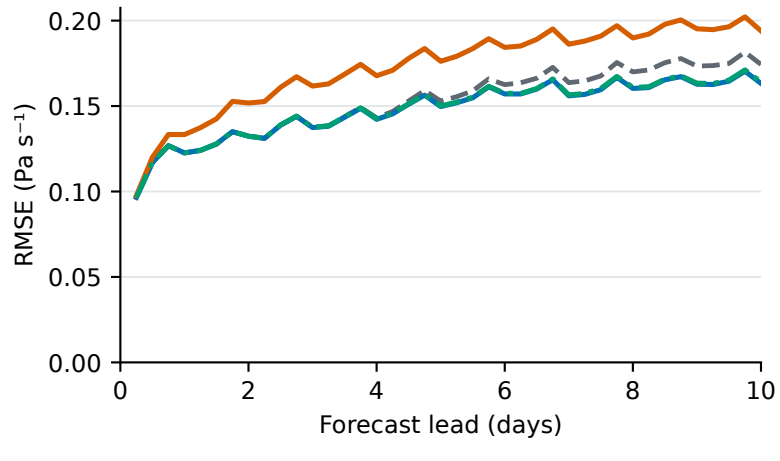
Meridional wind · all levels



Geopotential · all pressure levels



Vertical velocity · all levels



How to read these panels

Lower RMSE is better.
Each panel has its own physical units.

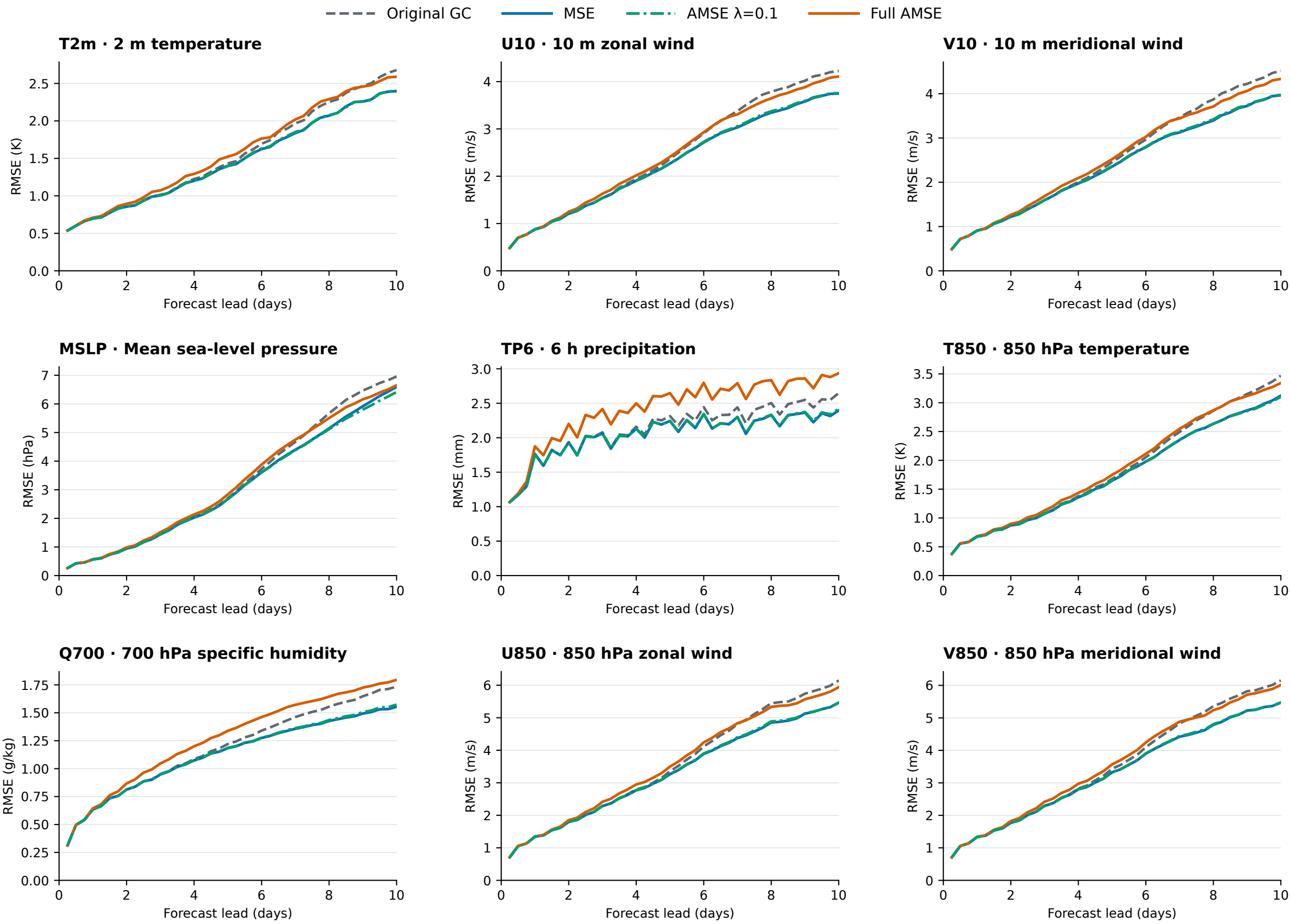
Upper-air groups pool squared error equally across all 13 pressure levels before taking the square root.

Global latitude-weighted errors pooled over the saved starts. These physical-variable RMSEs differ from the exact weighted GraphCast loss.

One common Original GC reference is shown; repeated baseline evaluations have small differences. No uncertainty intervals are available.

Global RMSE for the spectral diagnostic fields

8 matched 2023 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

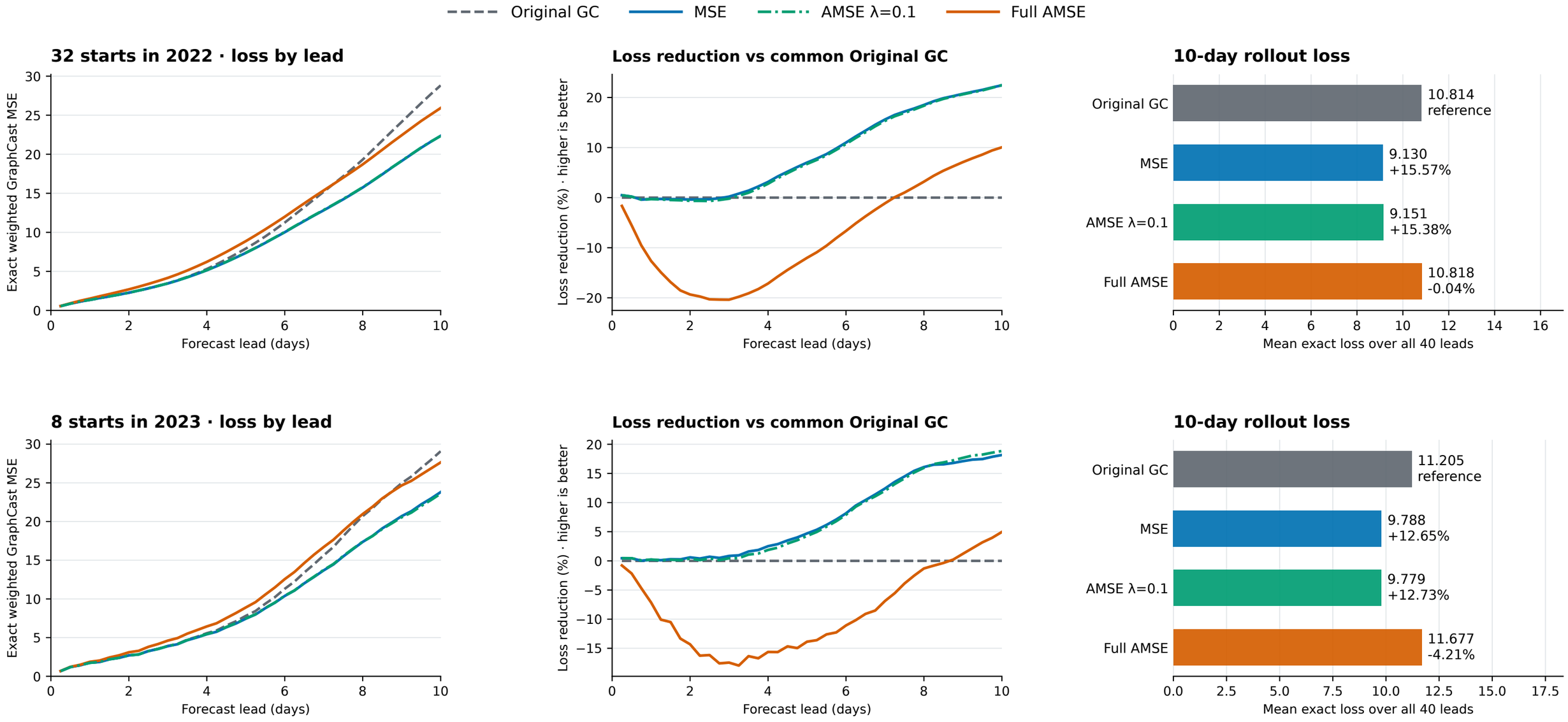


Global latitude-weighted physical RMSE at the specified level; the companion spectral curves use only 30–60° latitude in each hemisphere.

Global wind-speed RMSE (WS10 / WS850) was not saved; it cannot be reconstructed from component RMSEs. No uncertainty intervals.

Exact GraphCast forecast loss

Original variable weights, difference-standard-deviation normalization, pressure-level weights and latitude weights



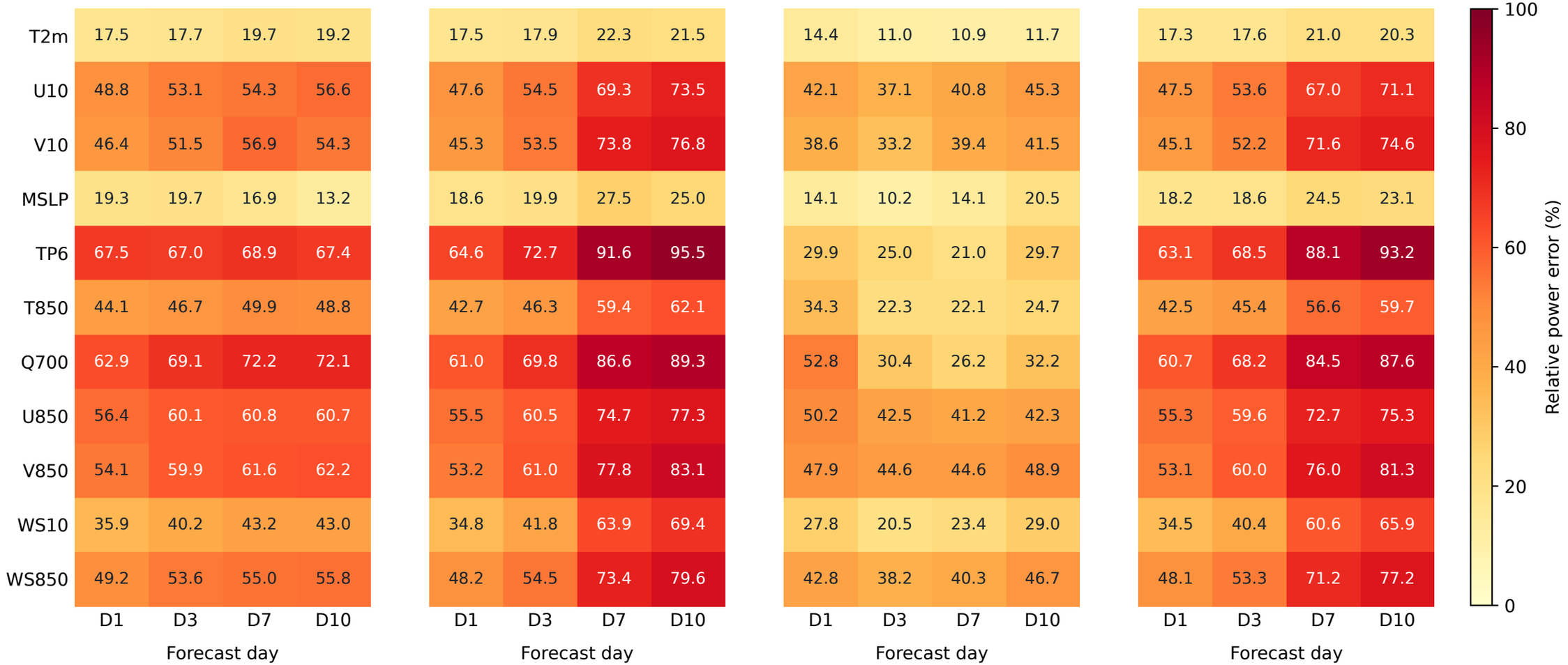
All plotted reductions use the same displayed Original GC within each dataset. They are ratios of losses; they are not RMSE reductions.

Rollout reduction is computed after pooling all 40 leads. Own-paired baseline results and baseline repeatability differences are retained in the data audit.

Spectral power error across all fields

Mean relative power error (%) · Lower is better

Each cell weights wavenumbers k = 1–180 equally. A common color scale is used for all models.



1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Field names and physical units are given on the individual diagnostic pages. No uncertainty intervals.

Forecast–truth coherence across all fields

Mean real spectral coherence · Higher is better

Each cell weights wavenumbers k = 1–180 equally. A common color scale is used for all models.



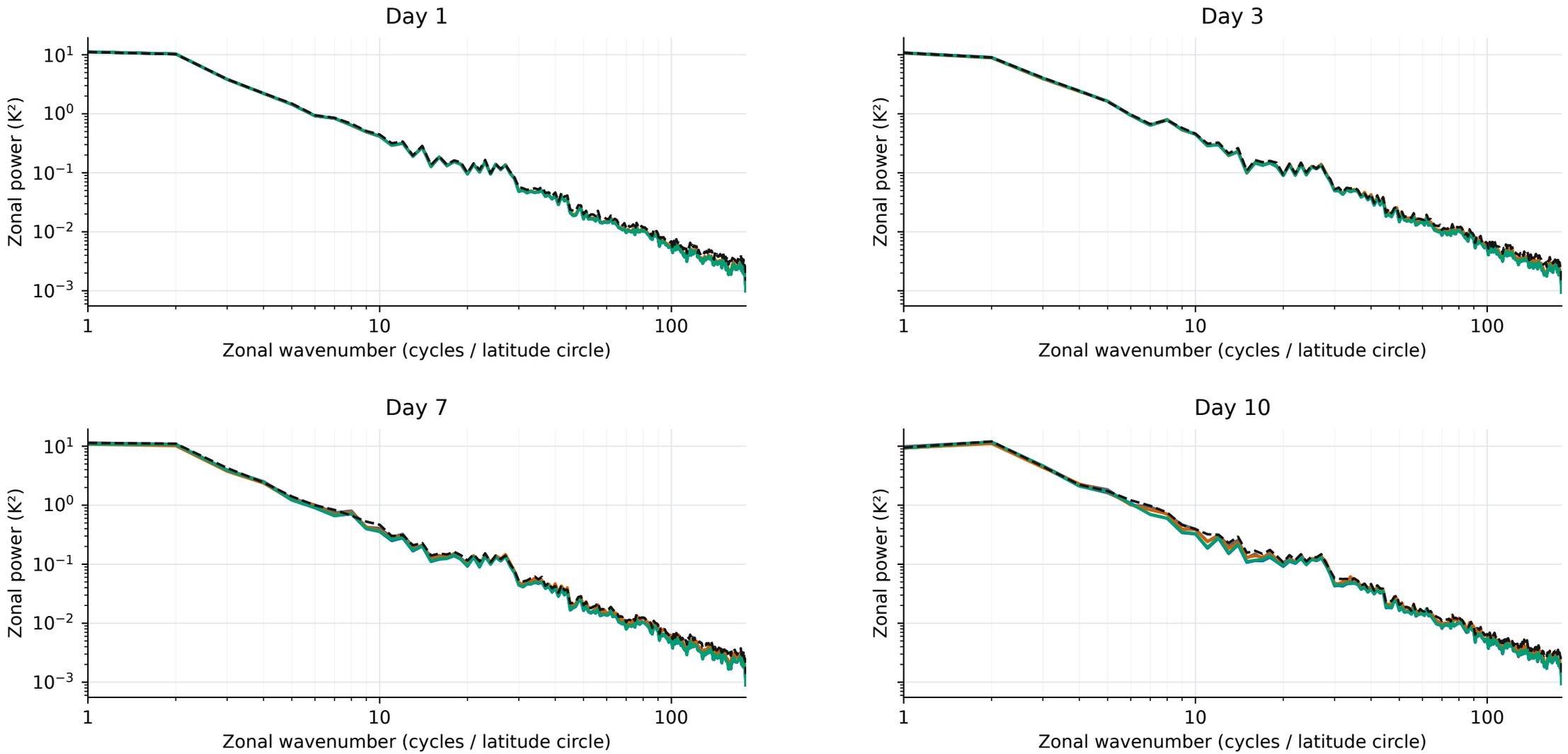
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Field names and physical units are given on the individual diagnostic pages. No uncertainty intervals.

2 m temperature (T2m; K)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



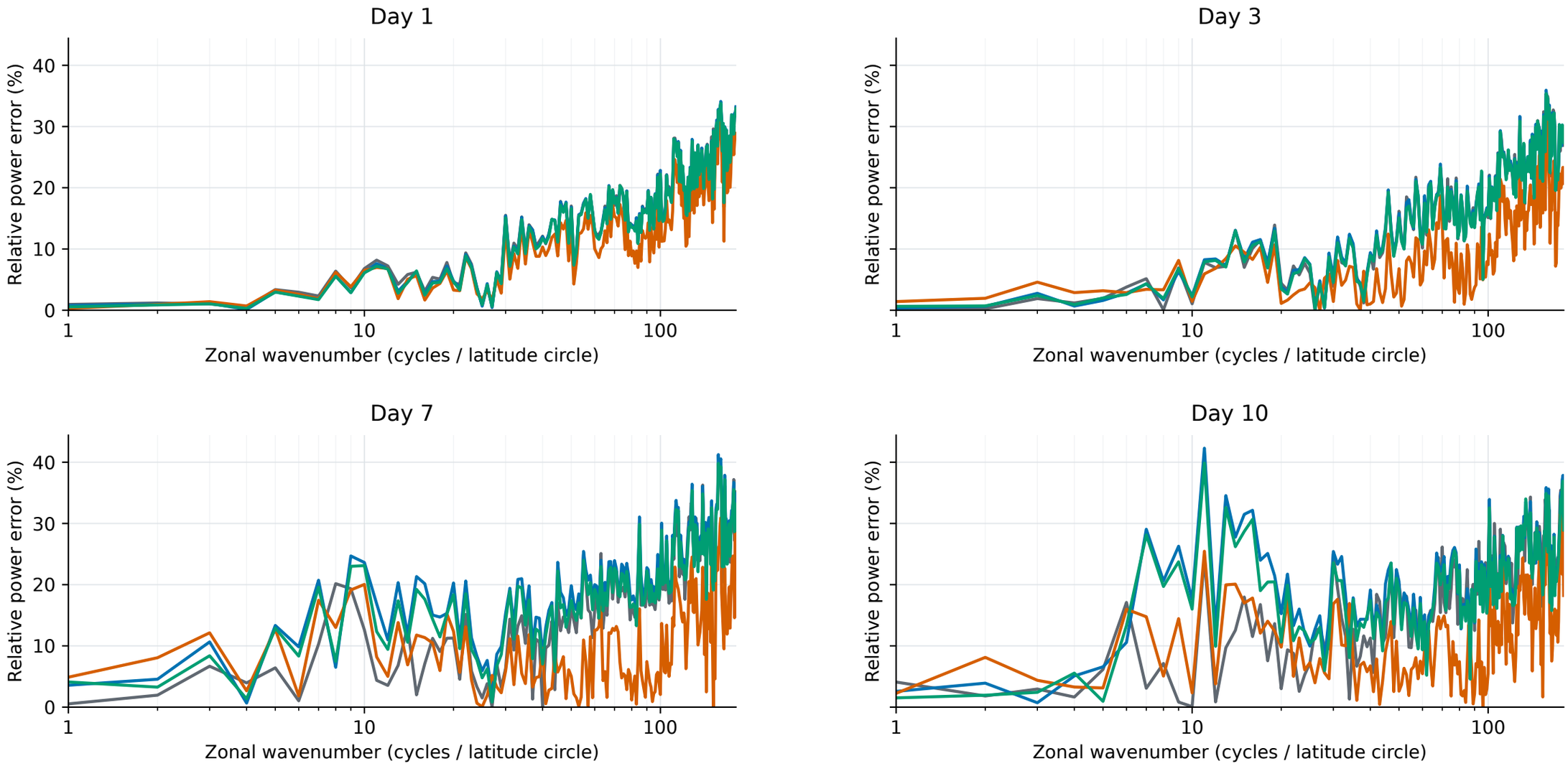
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

2 m temperature (T2m; K)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



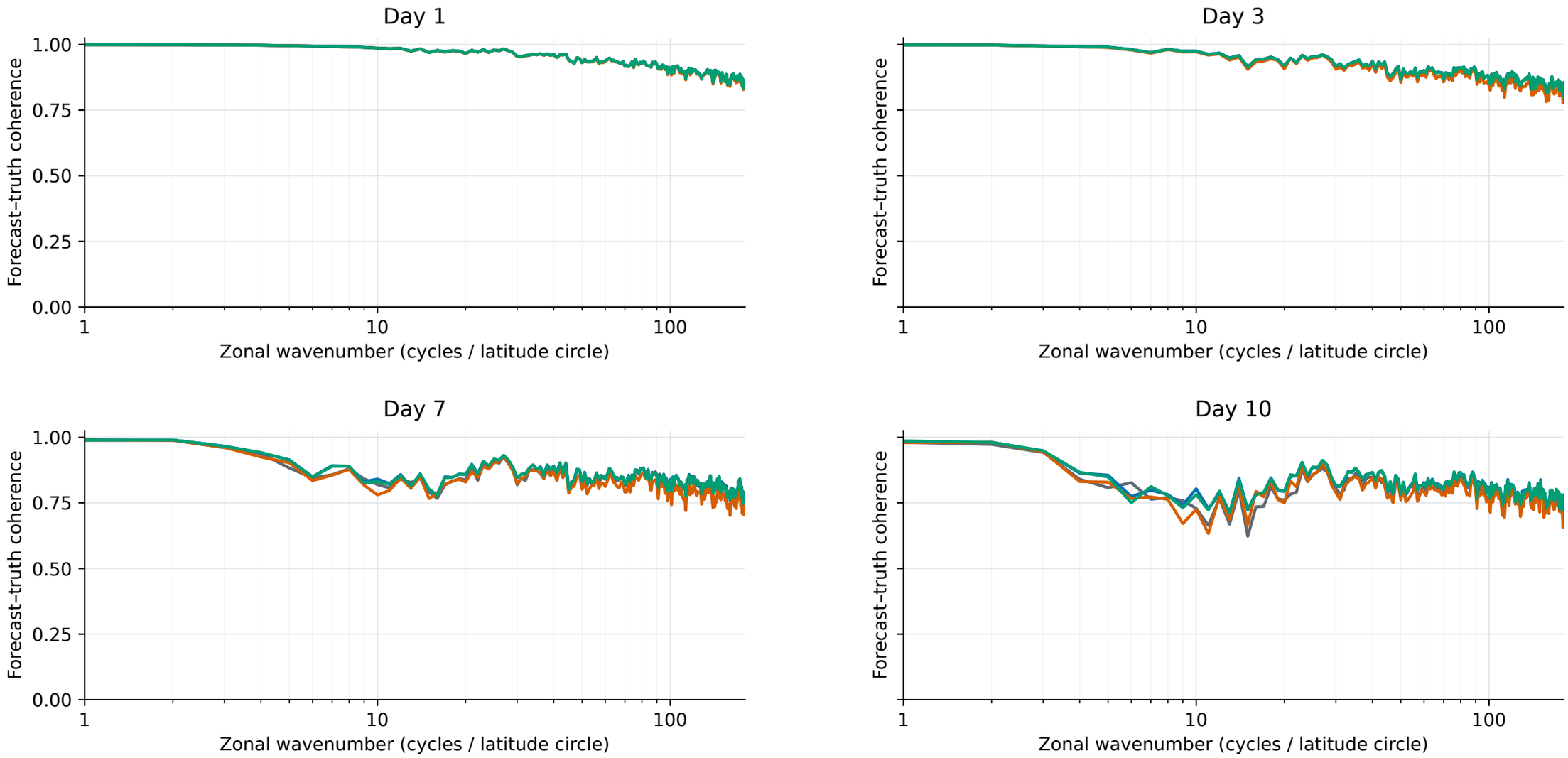
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

2 m temperature (T2m; K)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



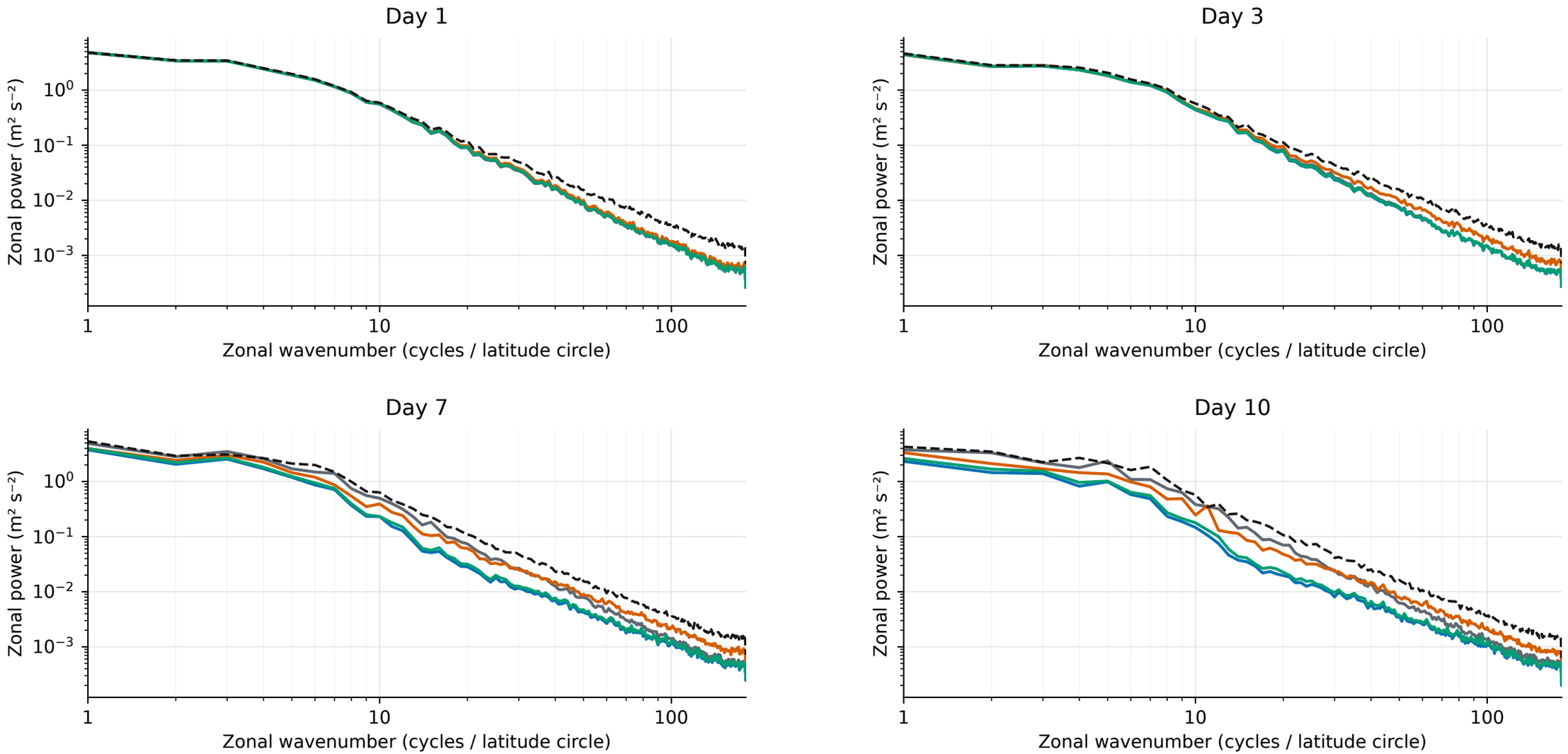
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m zonal wind (U10; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



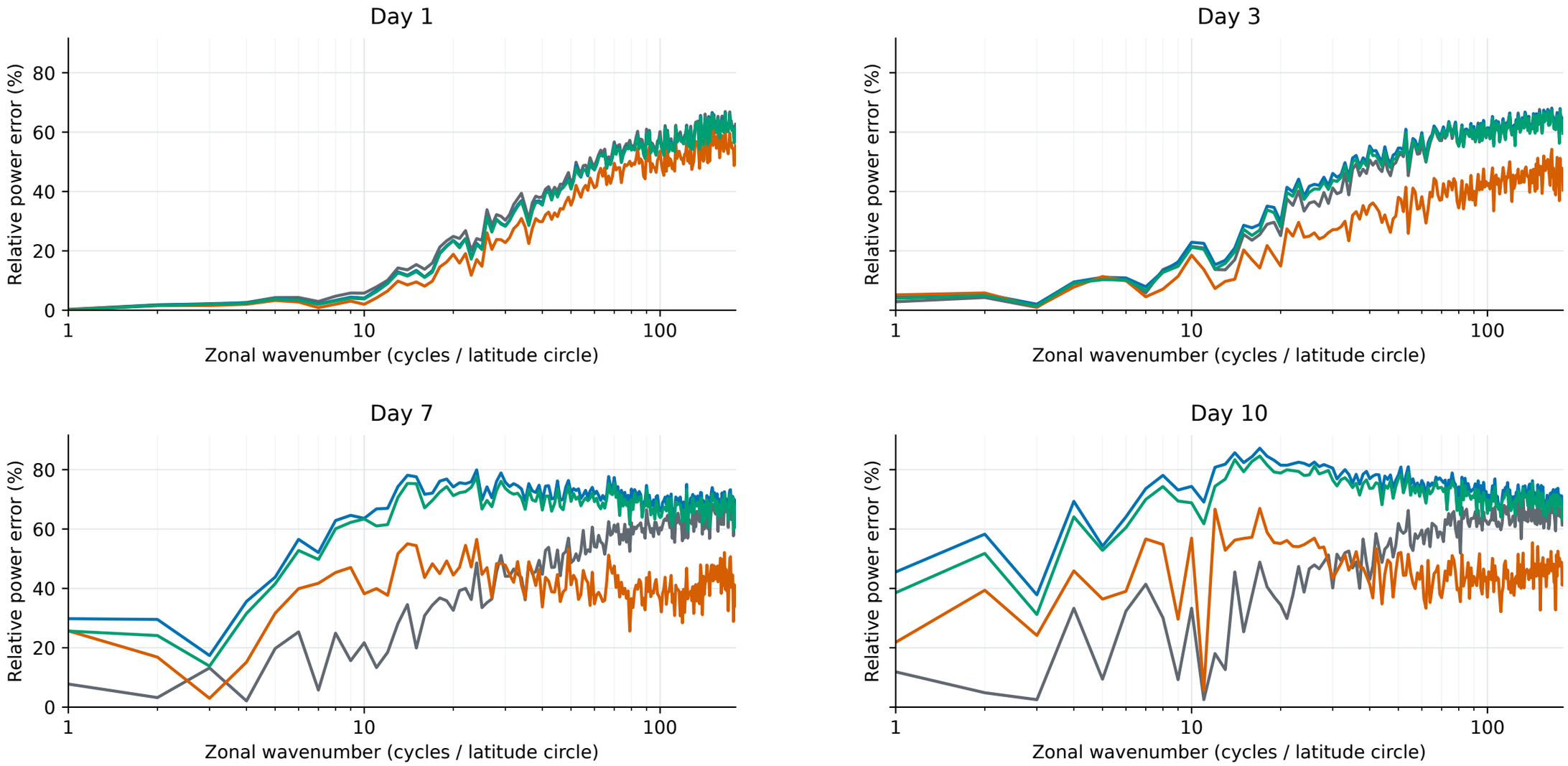
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m zonal wind (U10; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



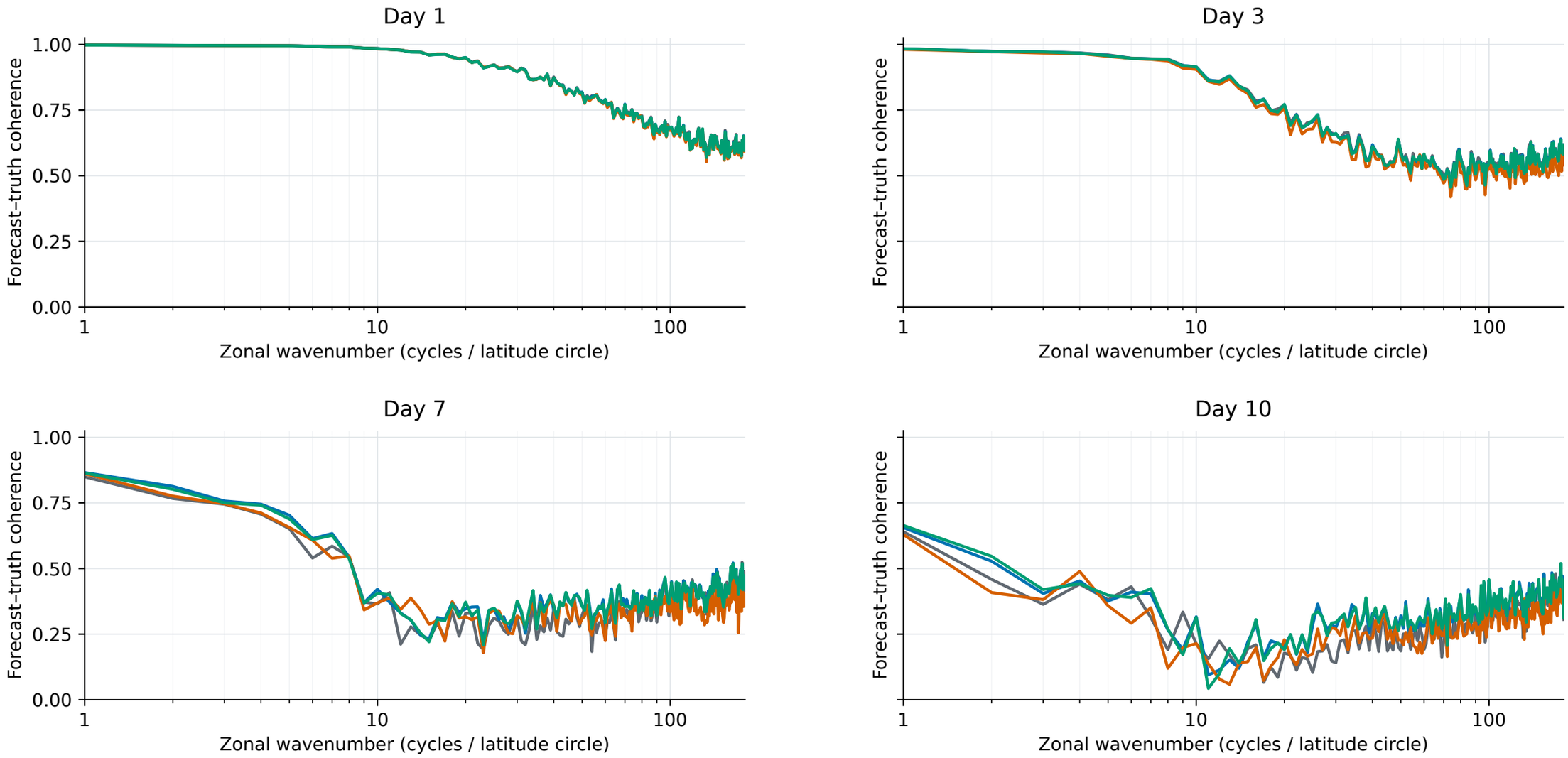
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m zonal wind (U10; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



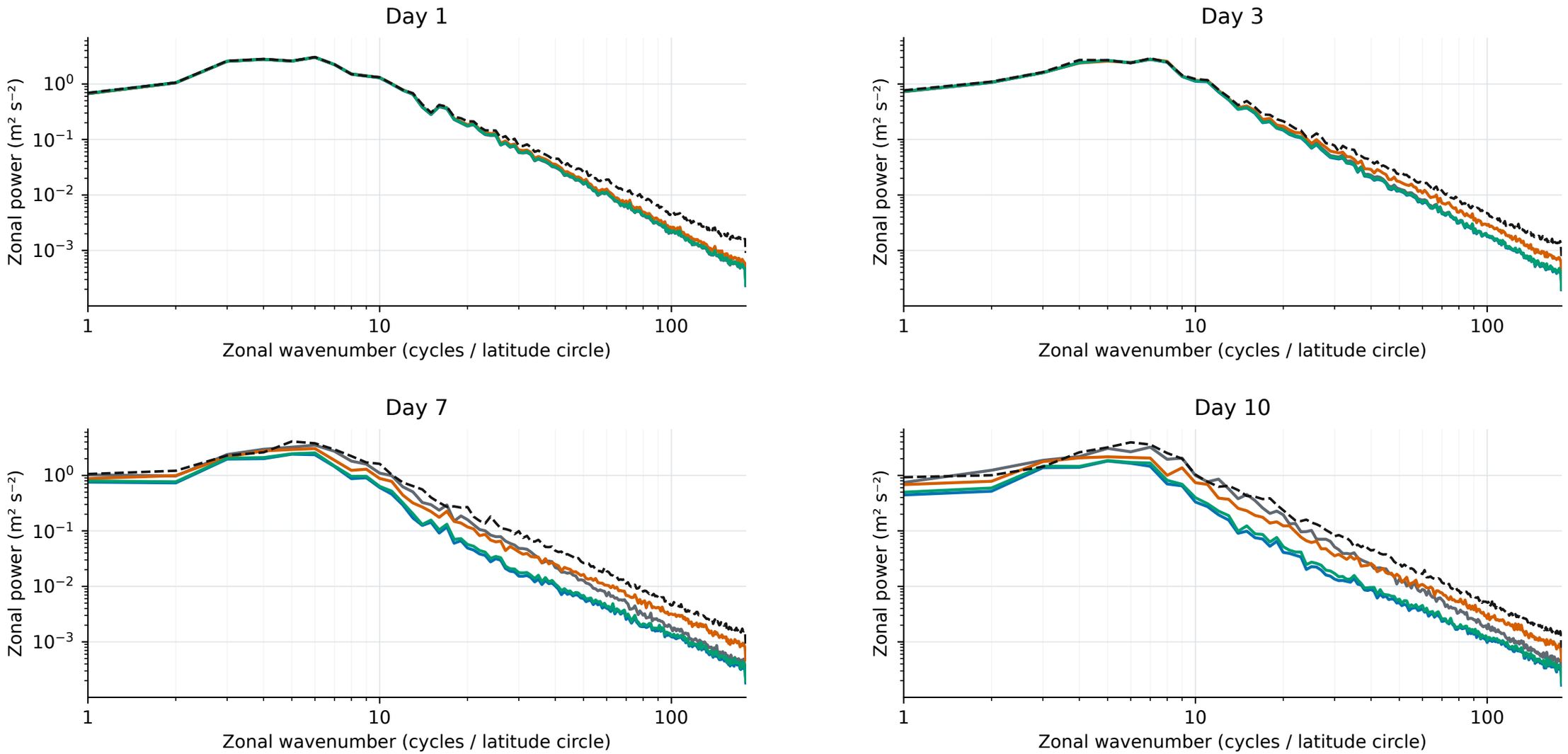
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m meridional wind (V10; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



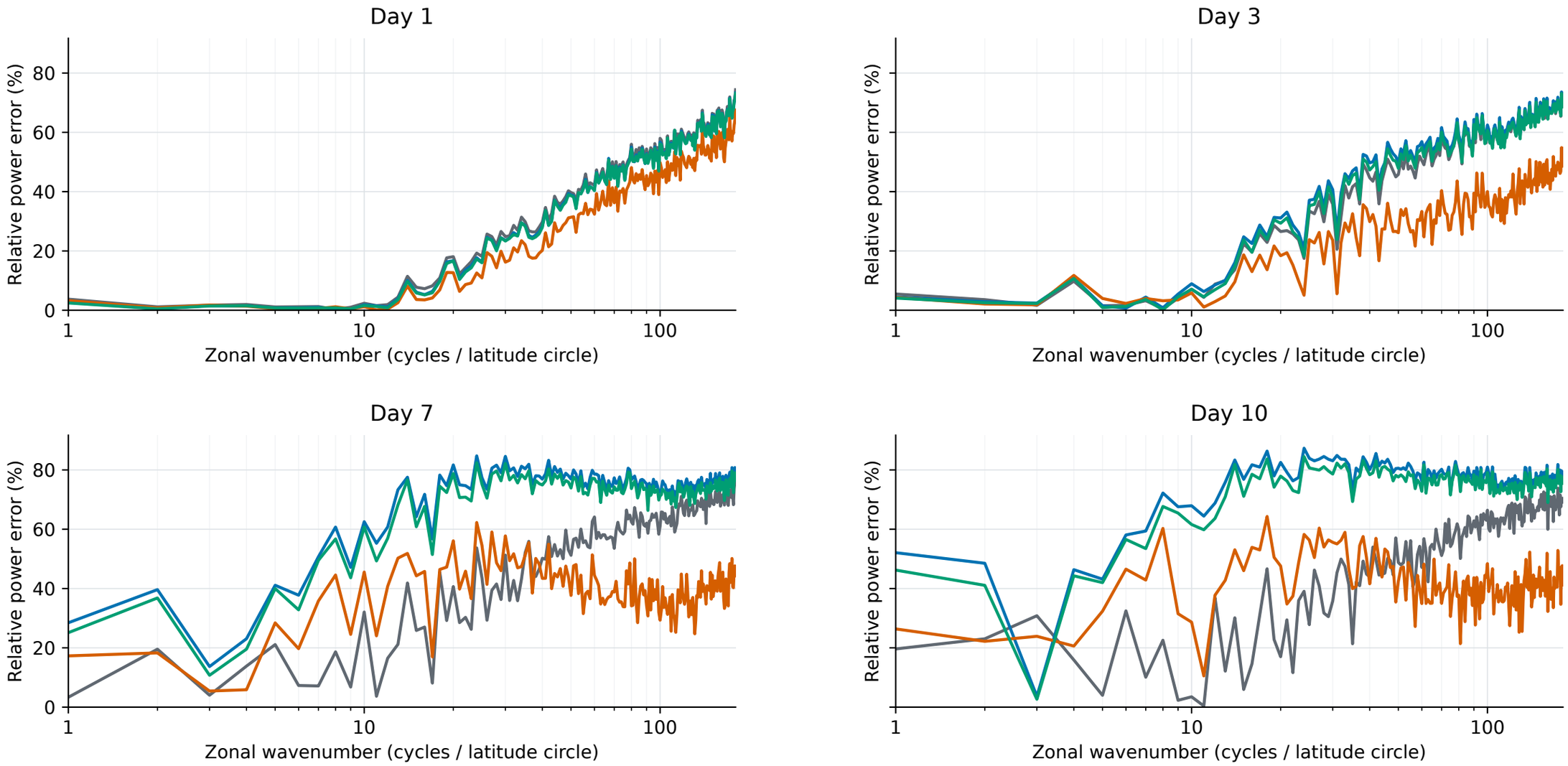
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m meridional wind (V10; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



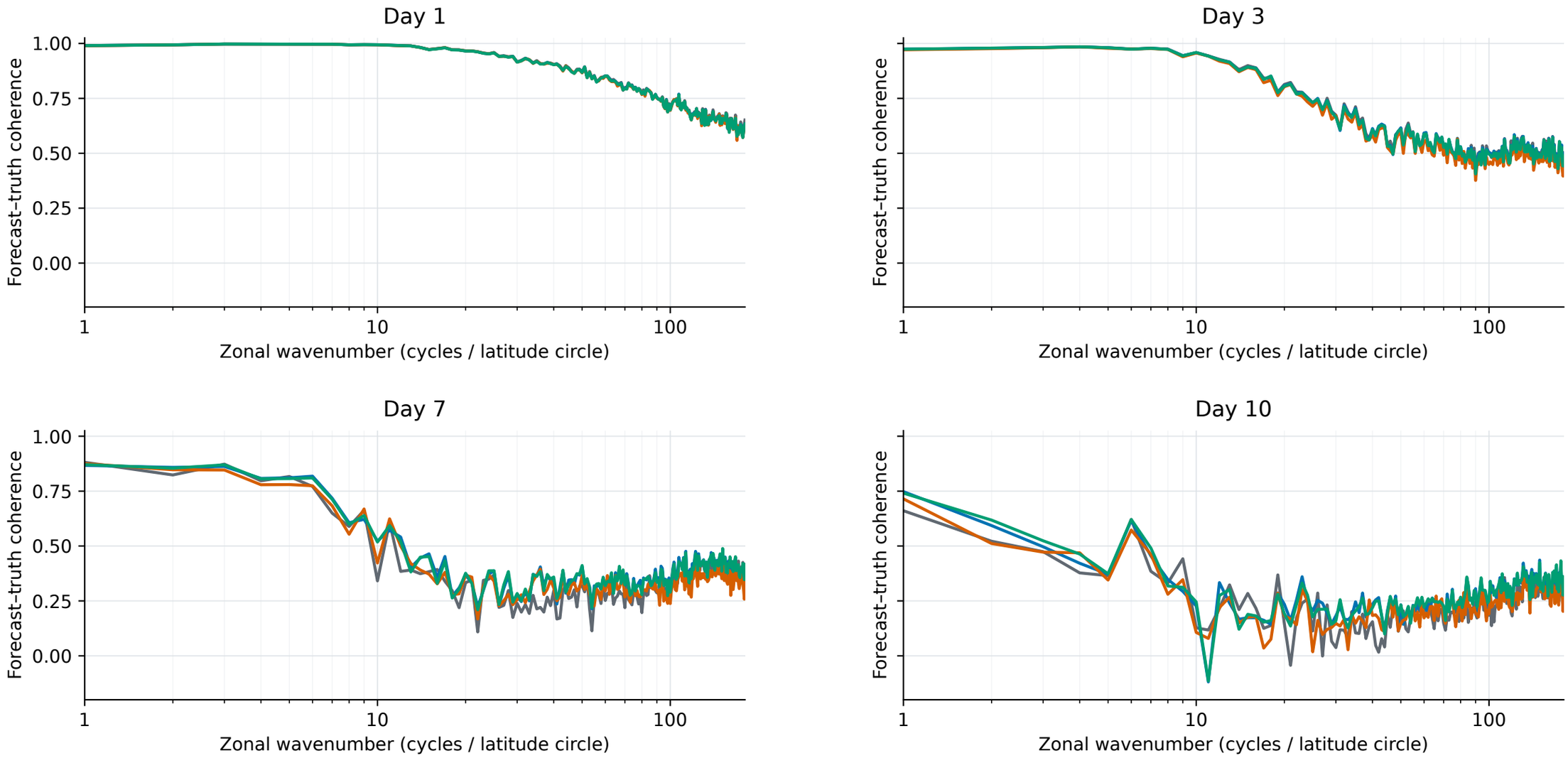
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m meridional wind (V10; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



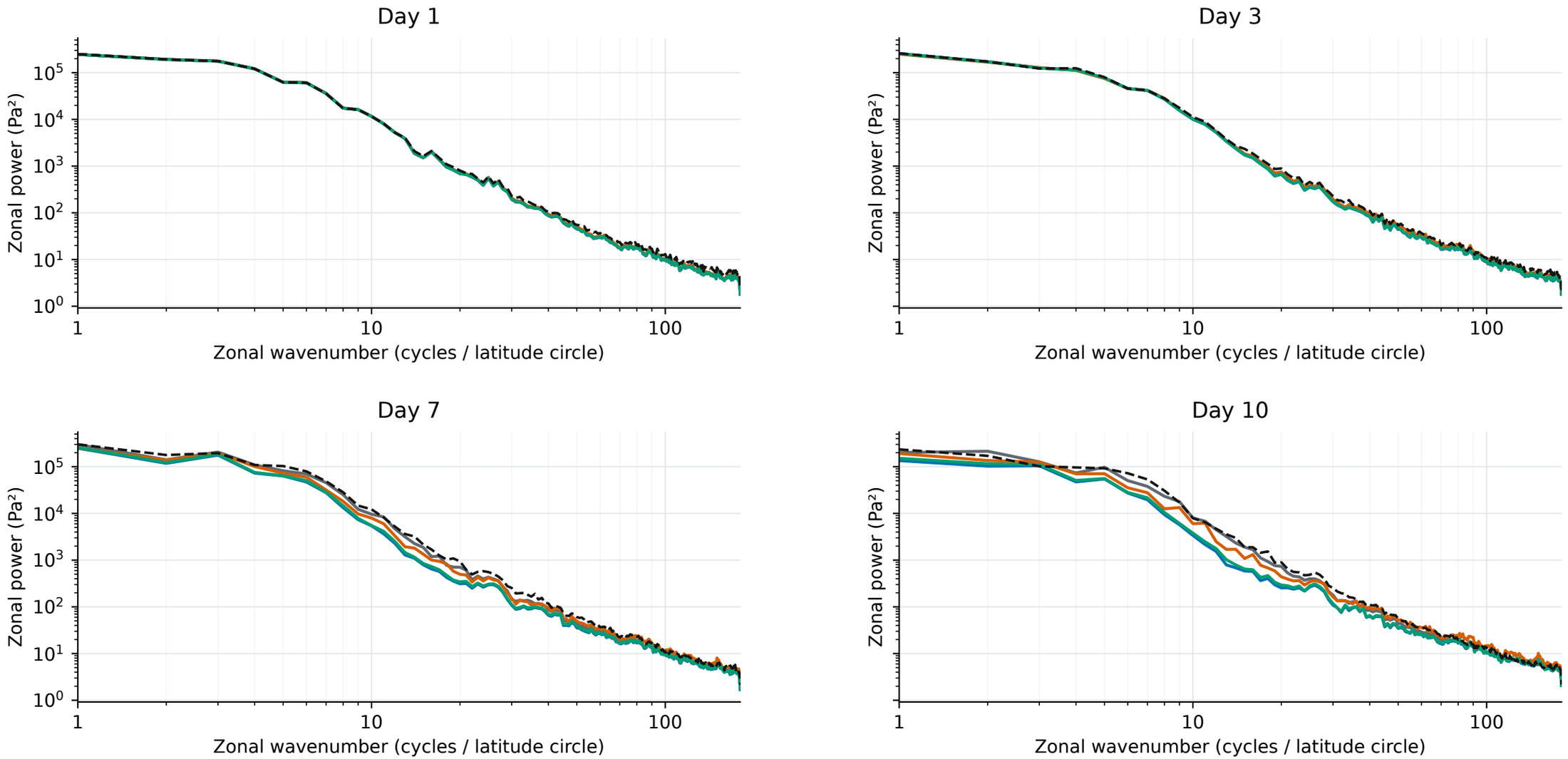
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Mean sea-level pressure (MSLP; Pa)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



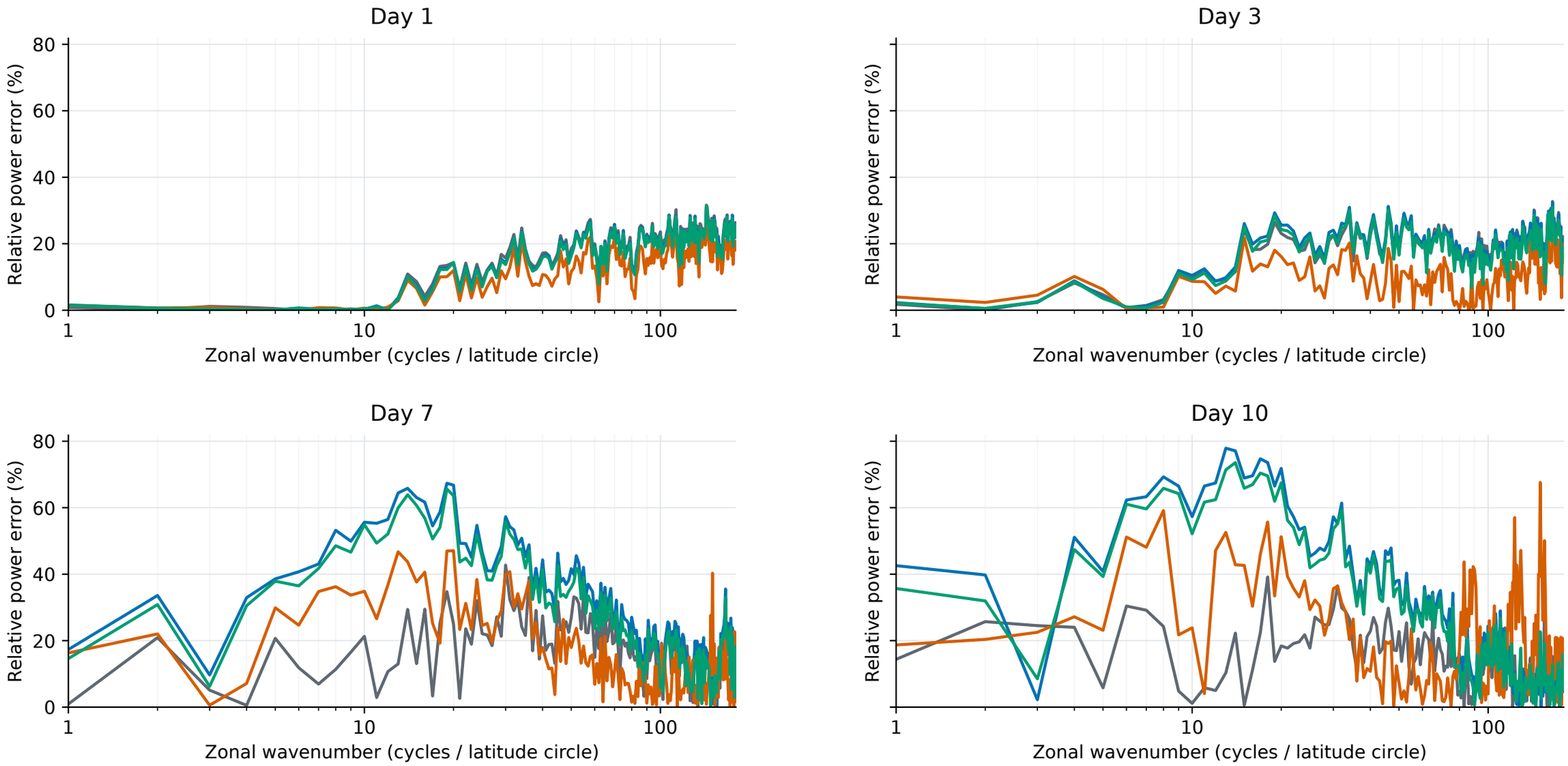
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Mean sea-level pressure (MSLP; Pa)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



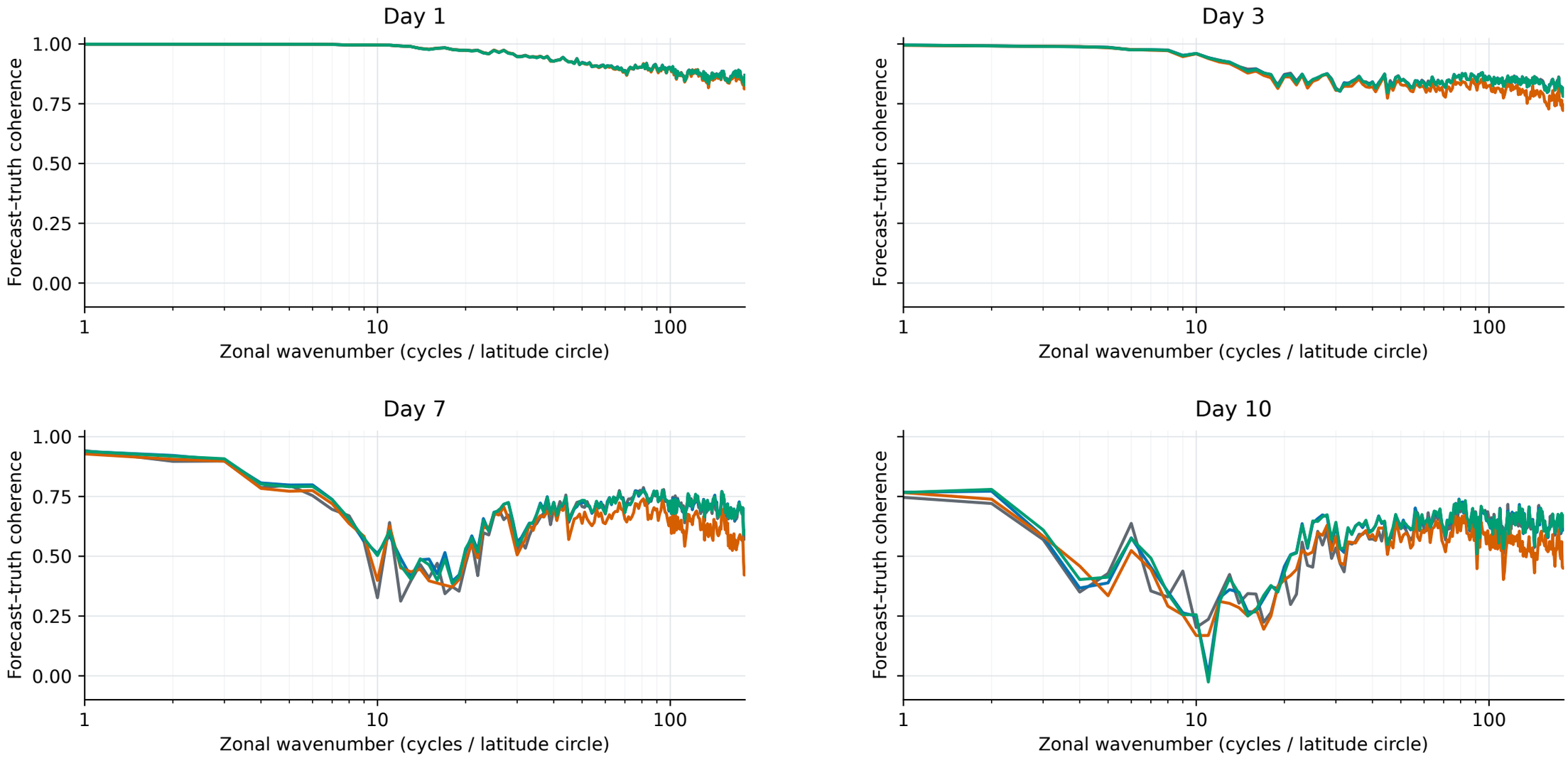
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Mean sea-level pressure (MSLP; Pa)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



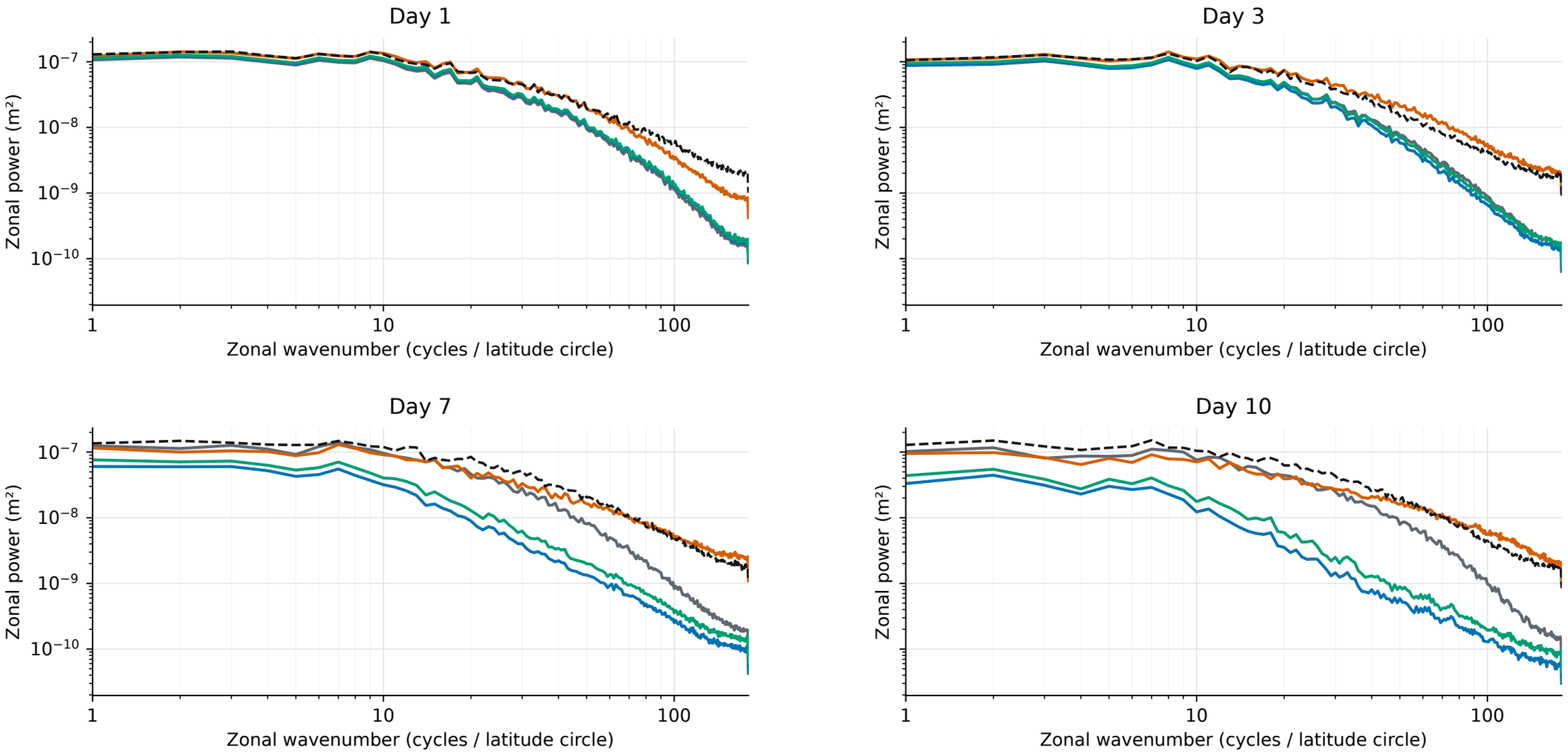
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

6-hour accumulated precipitation (TP6; m)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



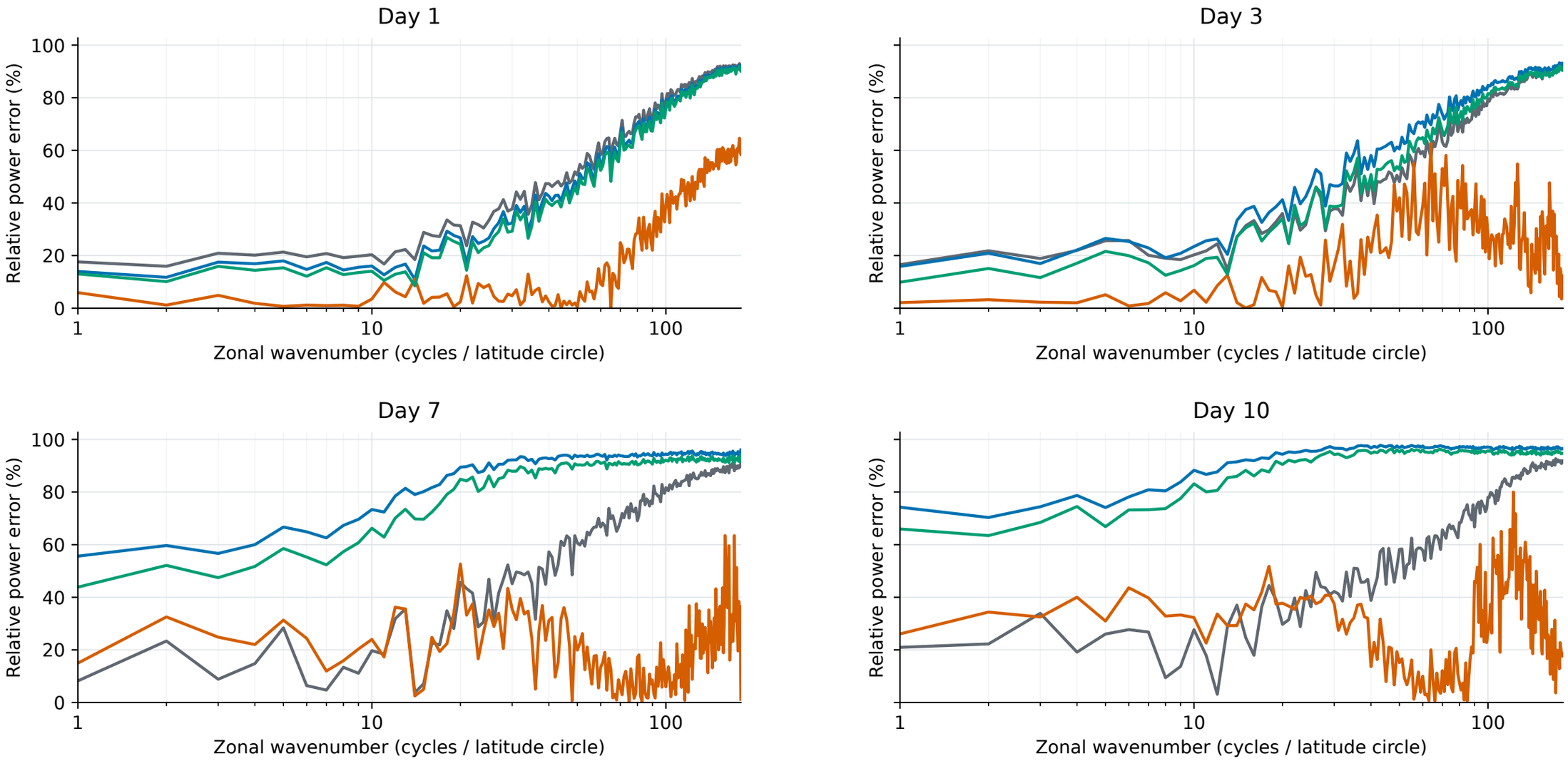
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

6-hour accumulated precipitation (TP6; m)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



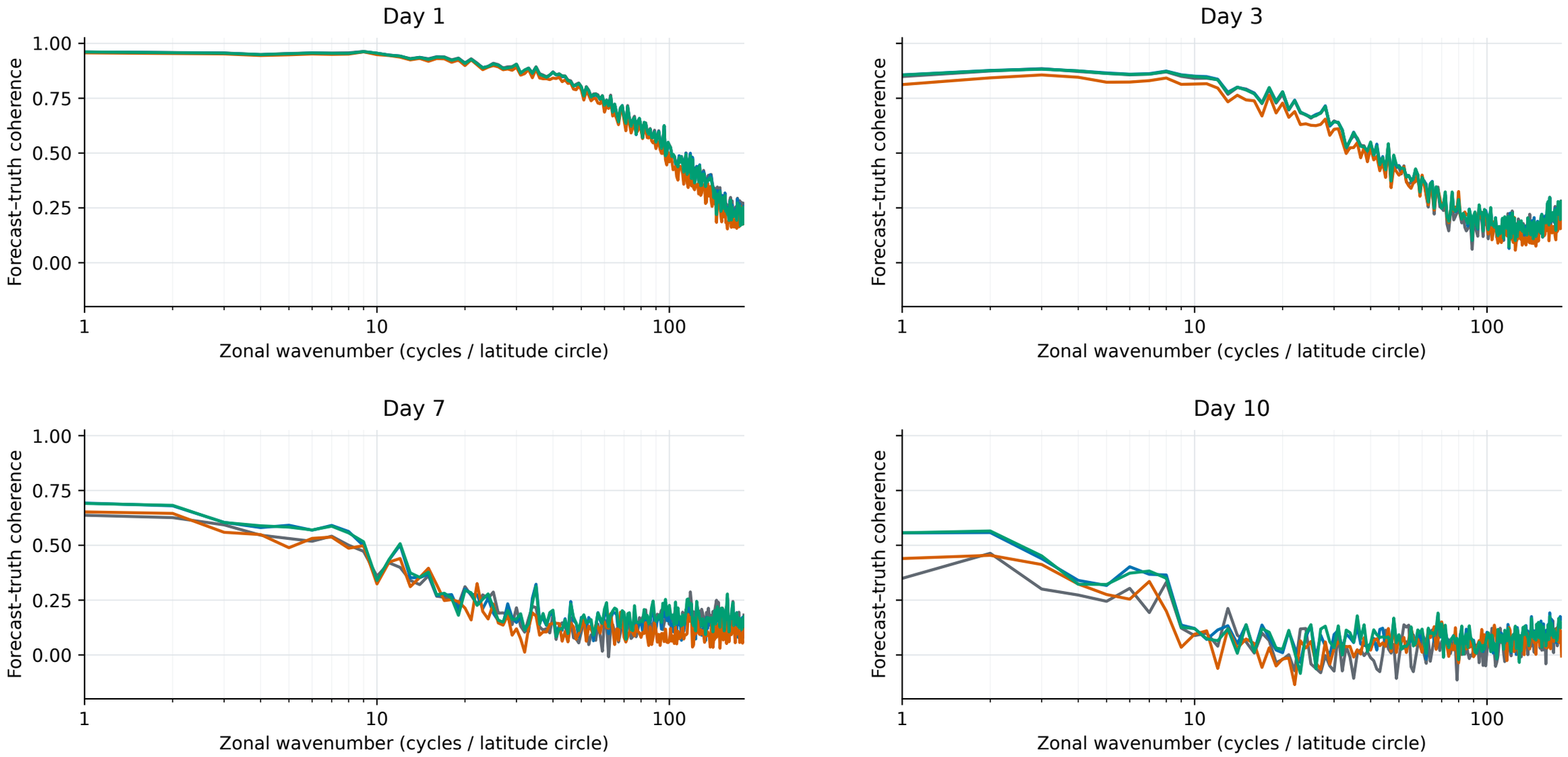
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

6-hour accumulated precipitation (TP6; m)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



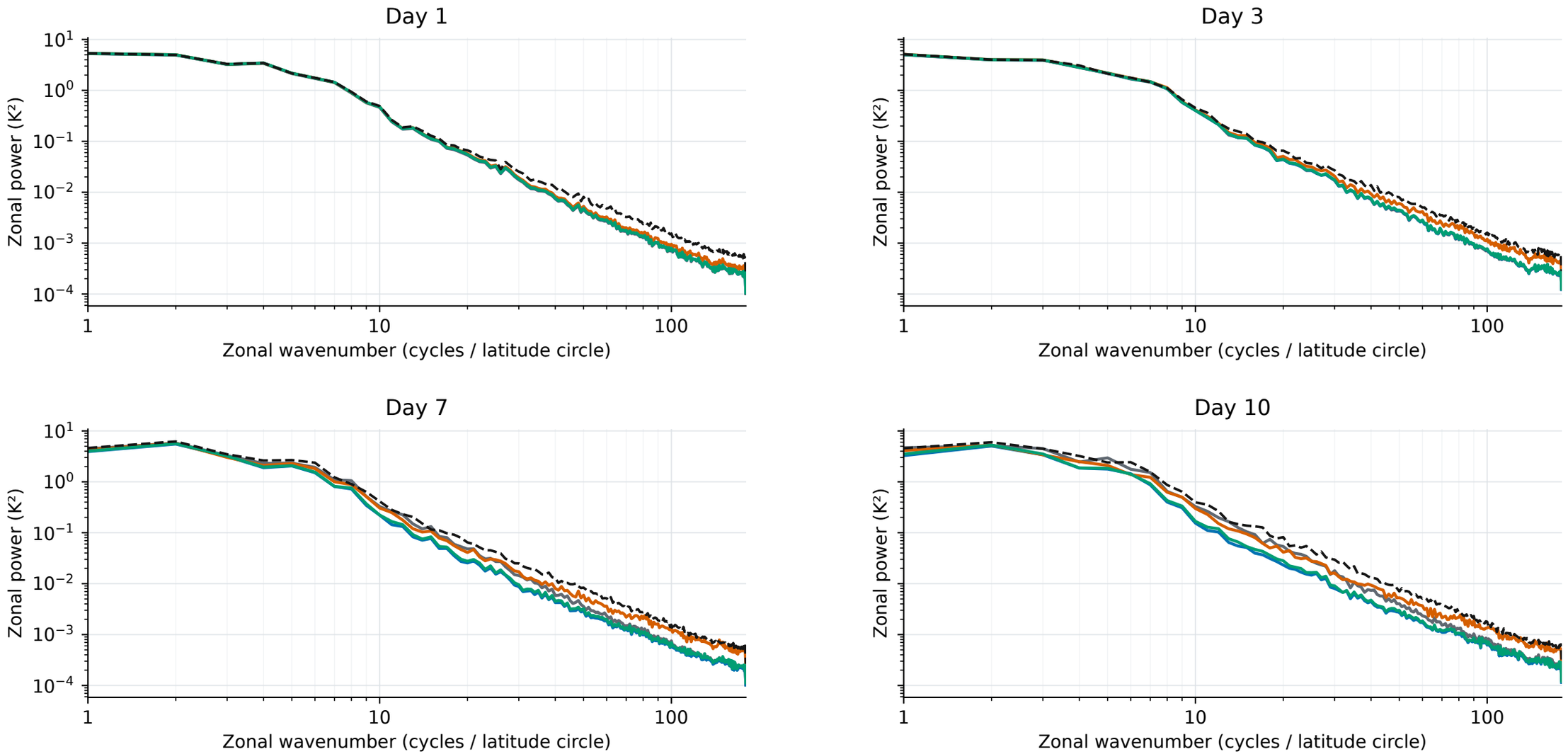
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Temperature at 850 hPa (T850; K)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



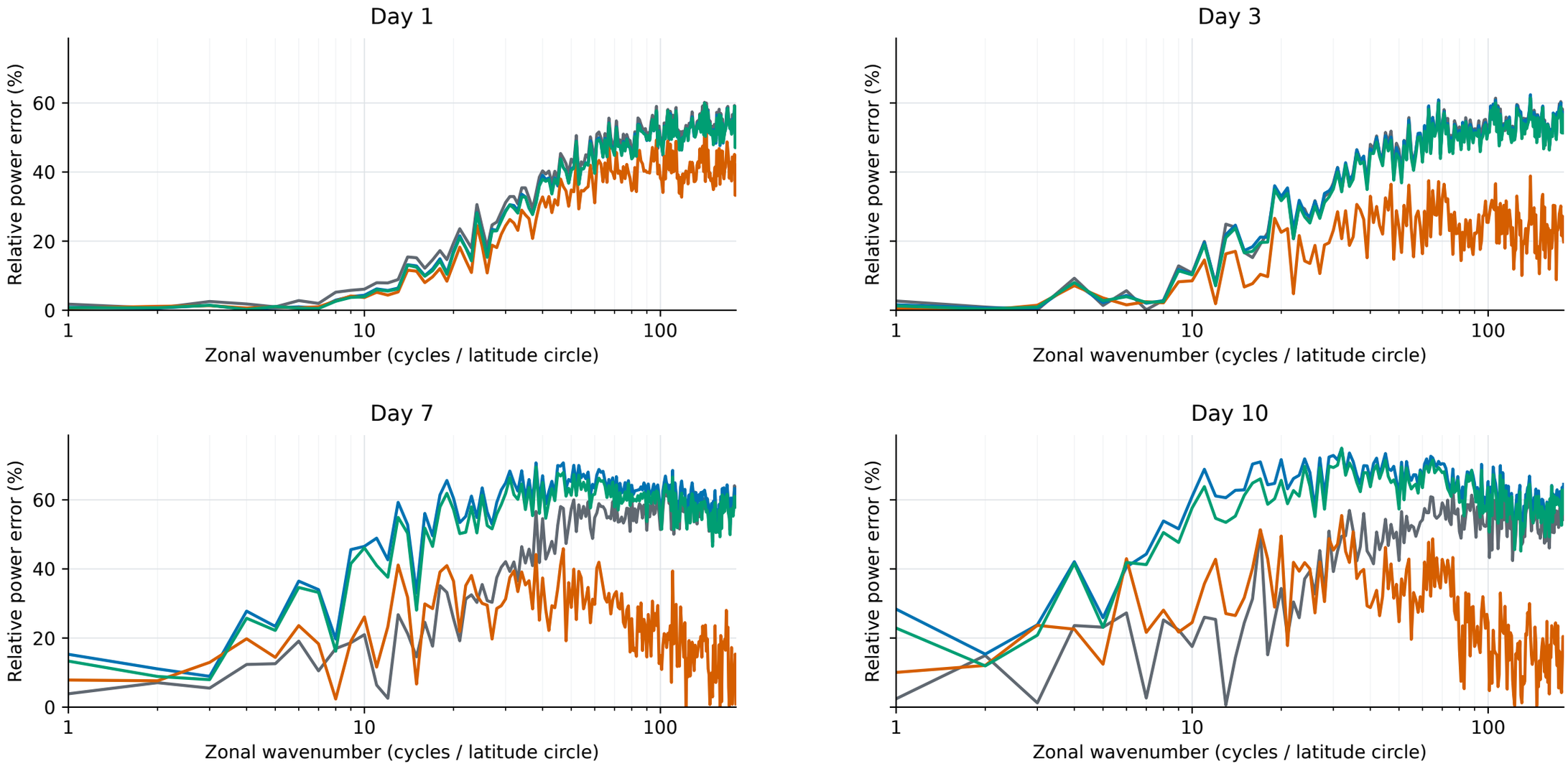
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Temperature at 850 hPa (T850; K)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



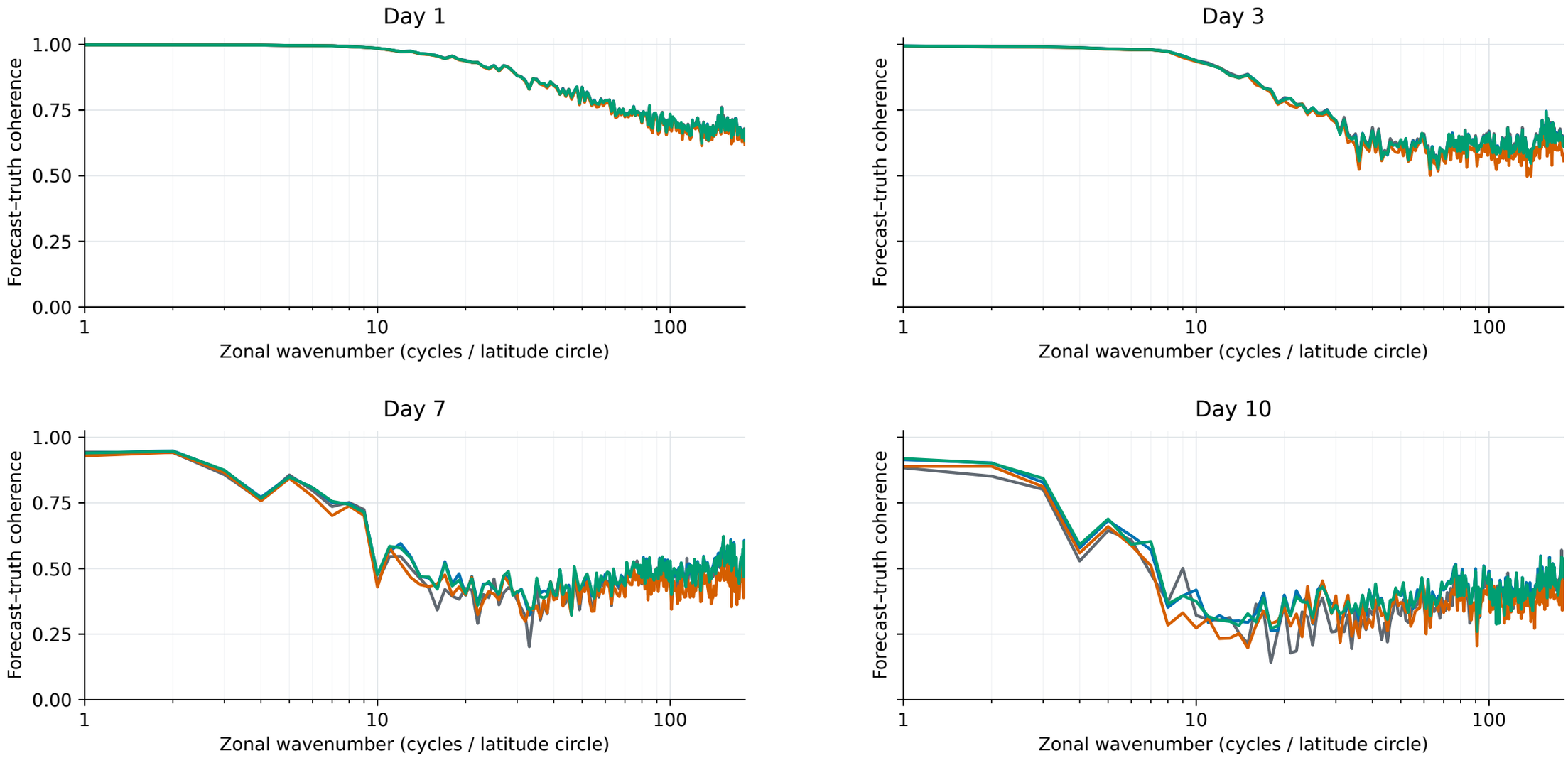
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Temperature at 850 hPa (T850; K)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



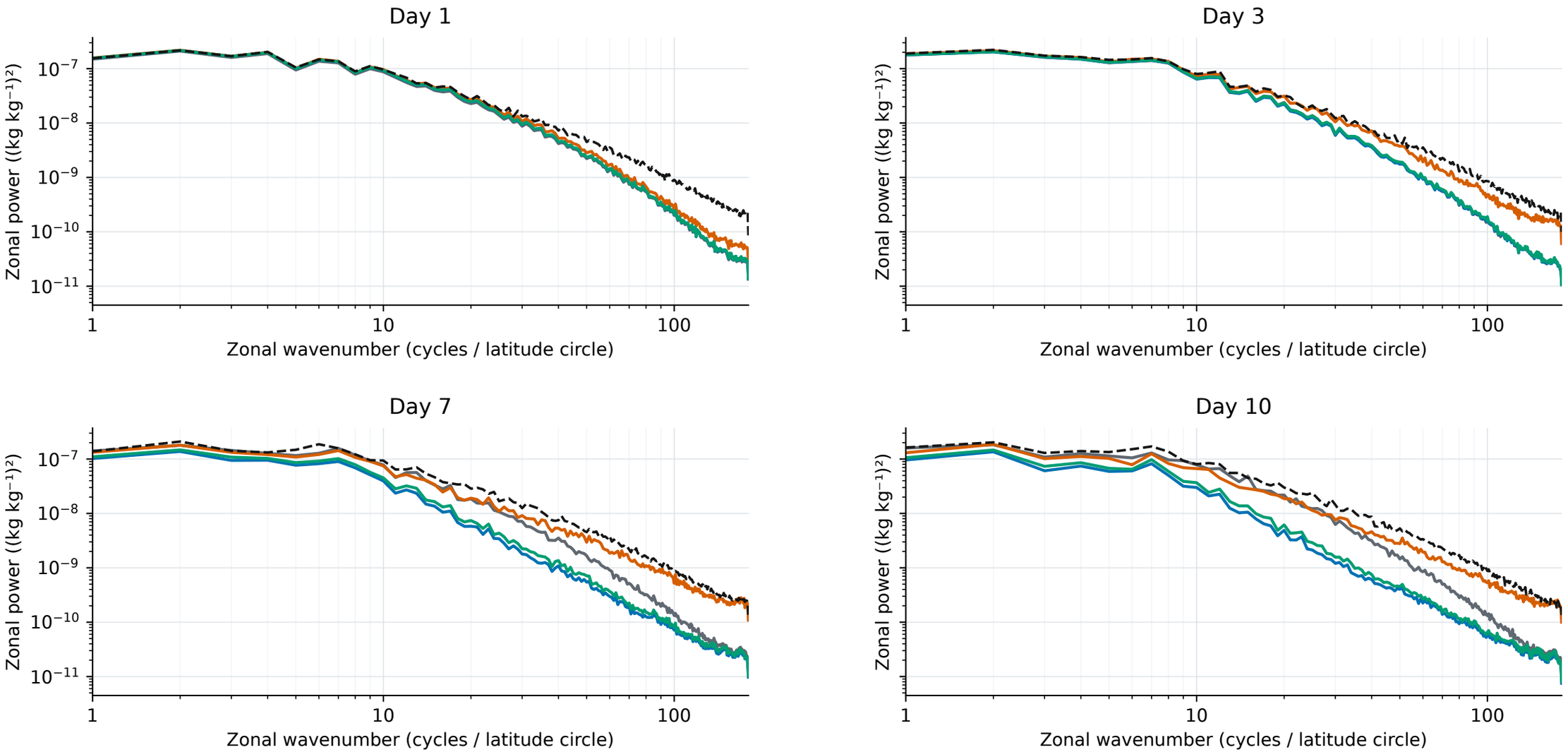
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Specific humidity at 700 hPa (Q700; kg kg⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



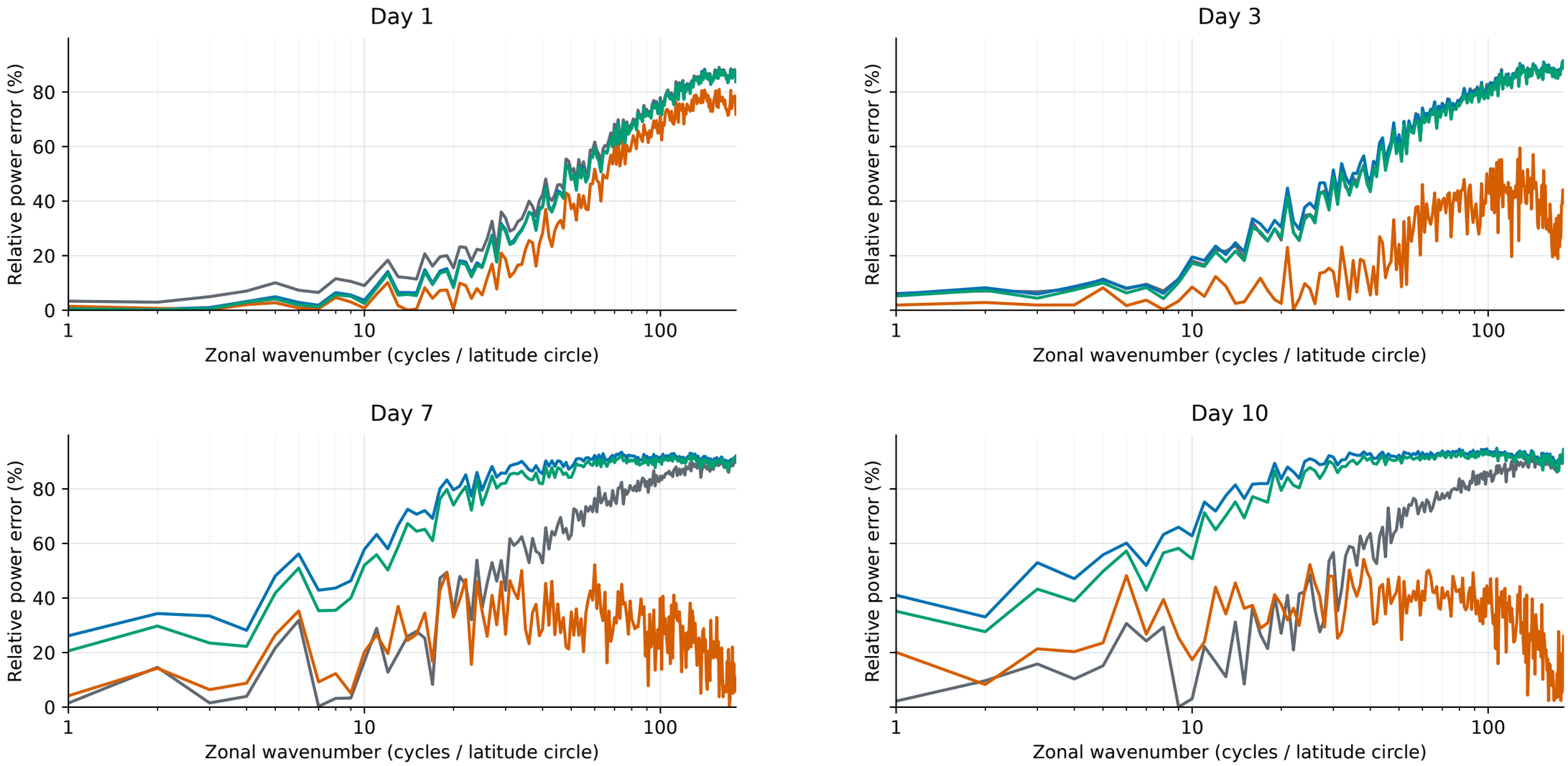
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Specific humidity at 700 hPa (Q700; kg kg⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



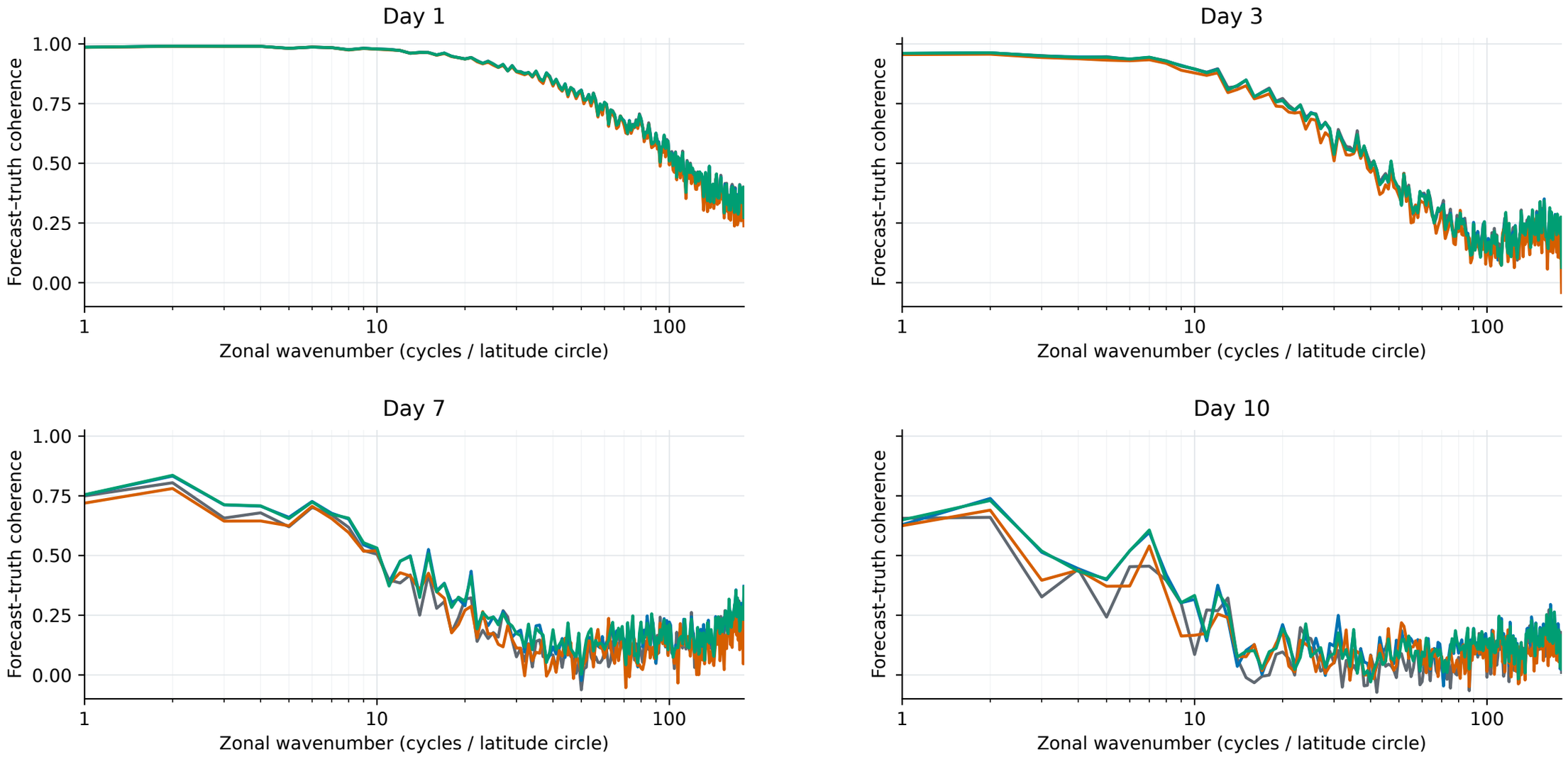
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Specific humidity at 700 hPa (Q700; kg kg⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



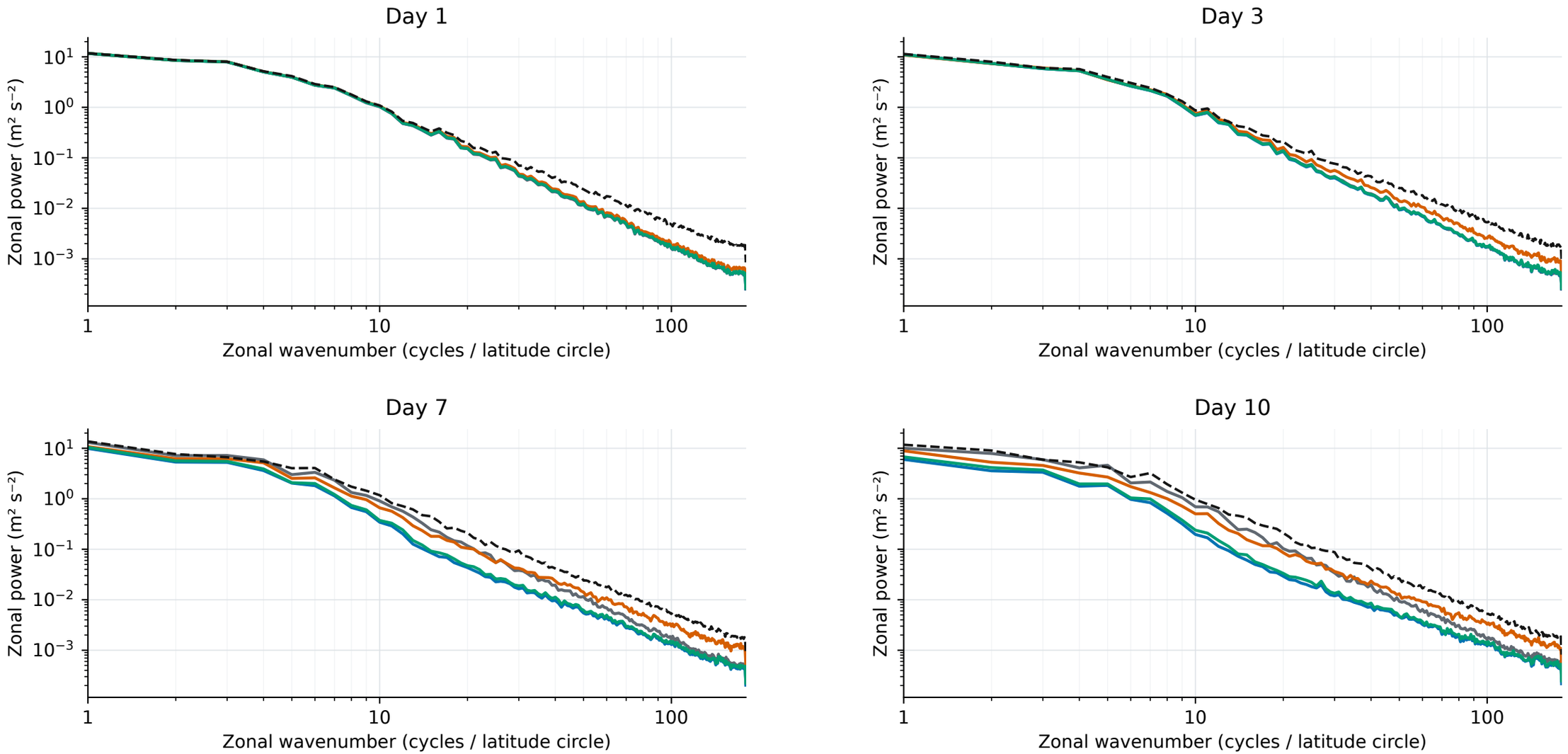
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Zonal wind at 850 hPa (U850; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



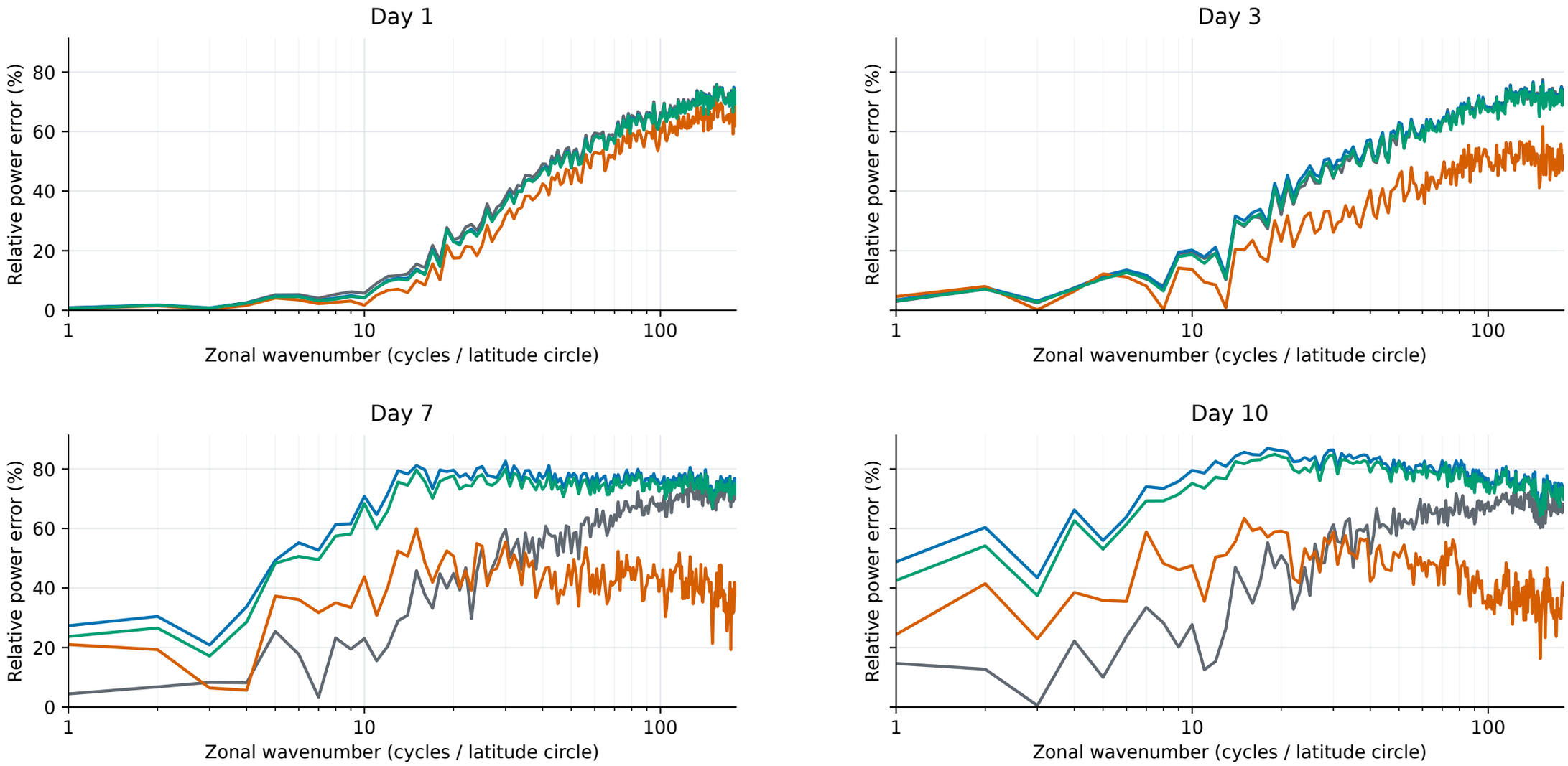
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Zonal wind at 850 hPa (U850; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



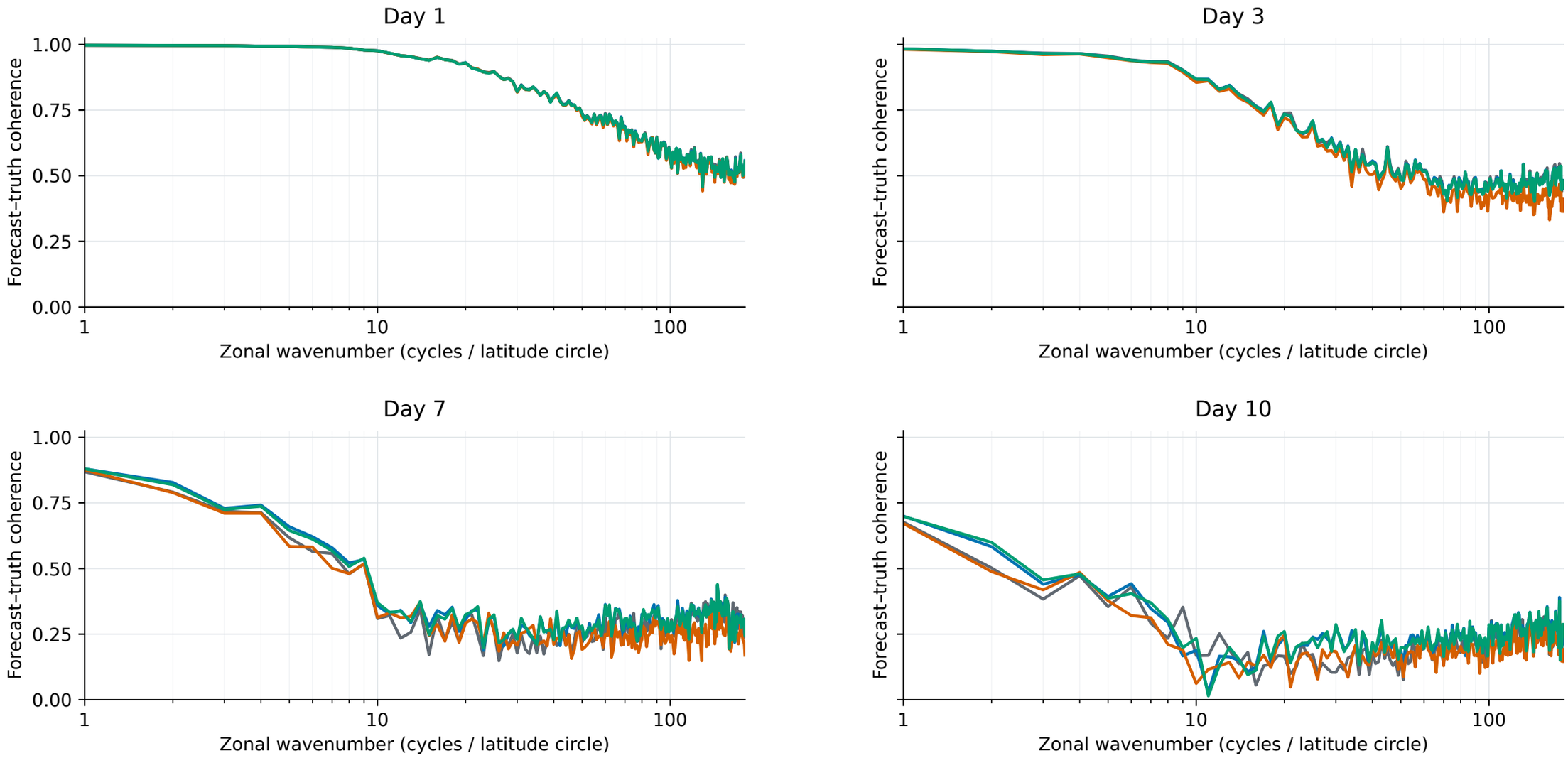
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Zonal wind at 850 hPa (U850; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



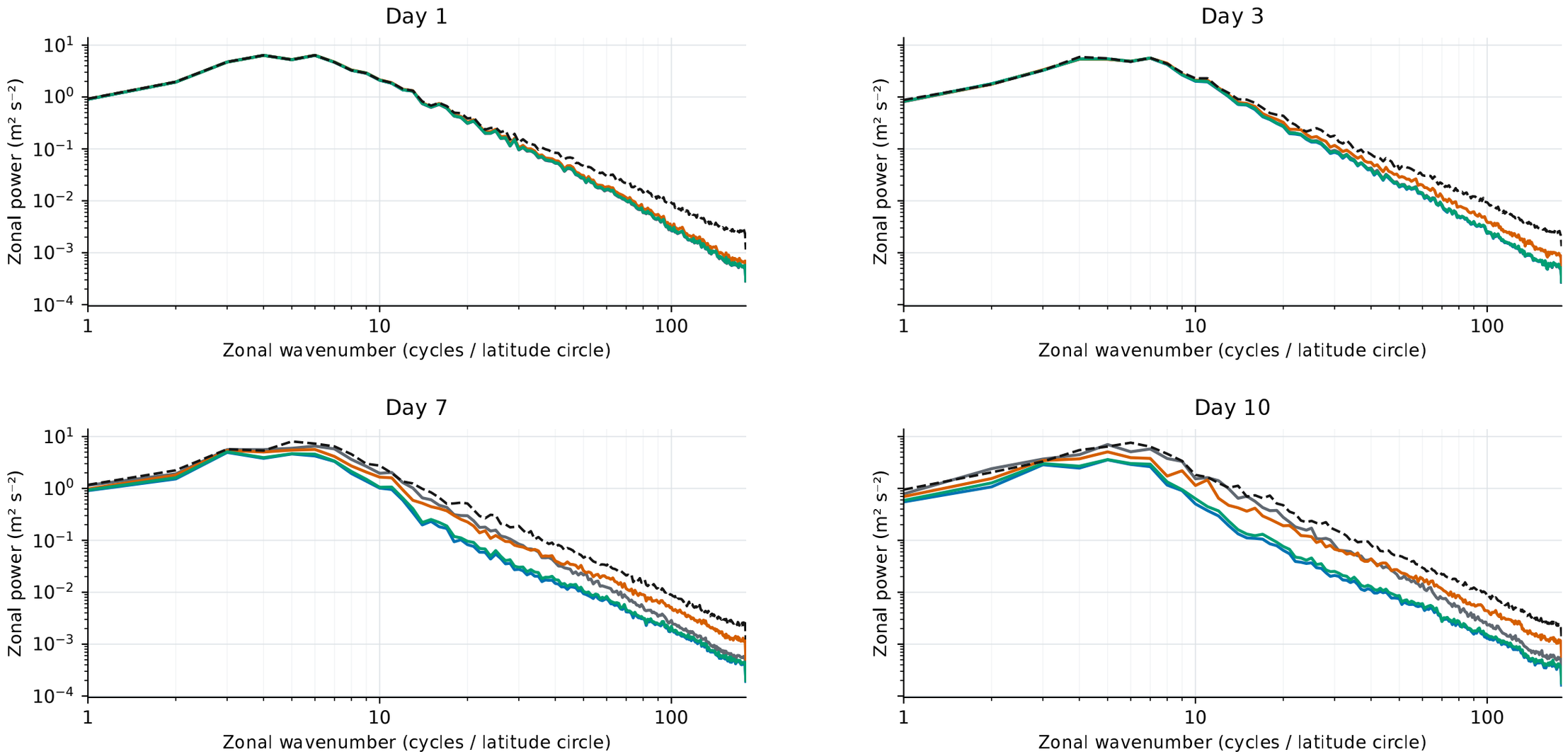
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Meridional wind at 850 hPa (V850; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



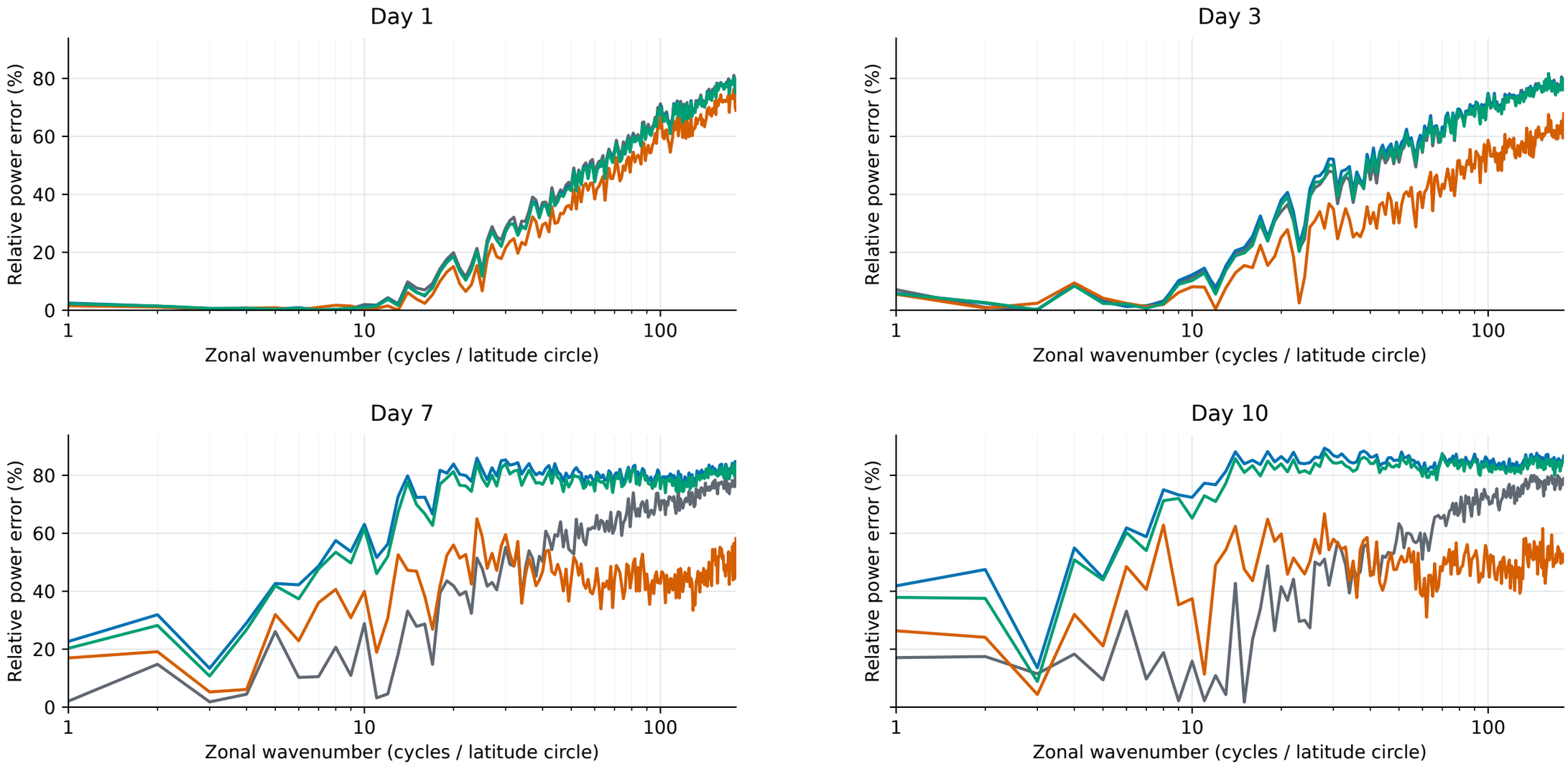
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Meridional wind at 850 hPa (V850; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



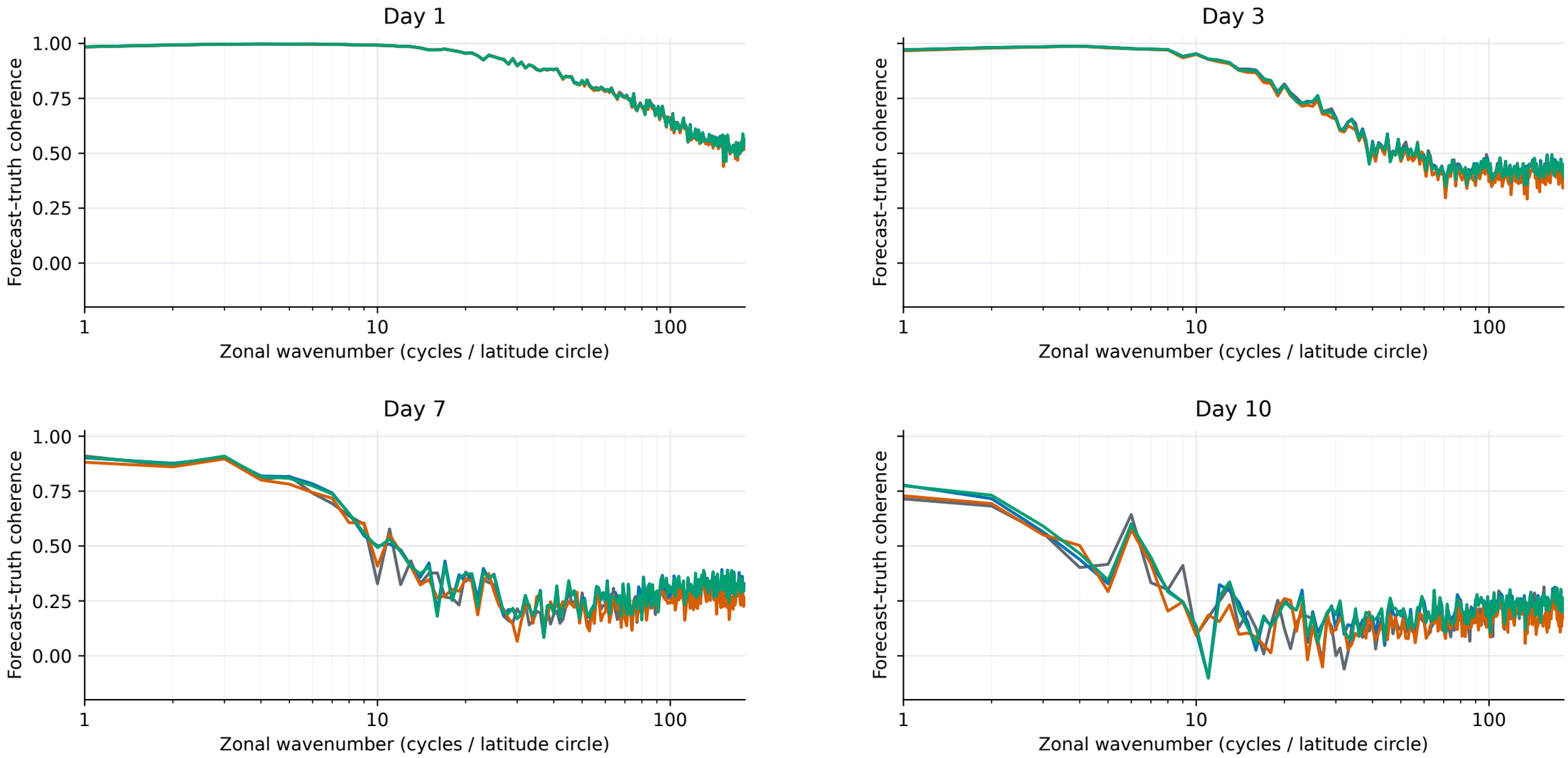
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Meridional wind at 850 hPa (V850; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



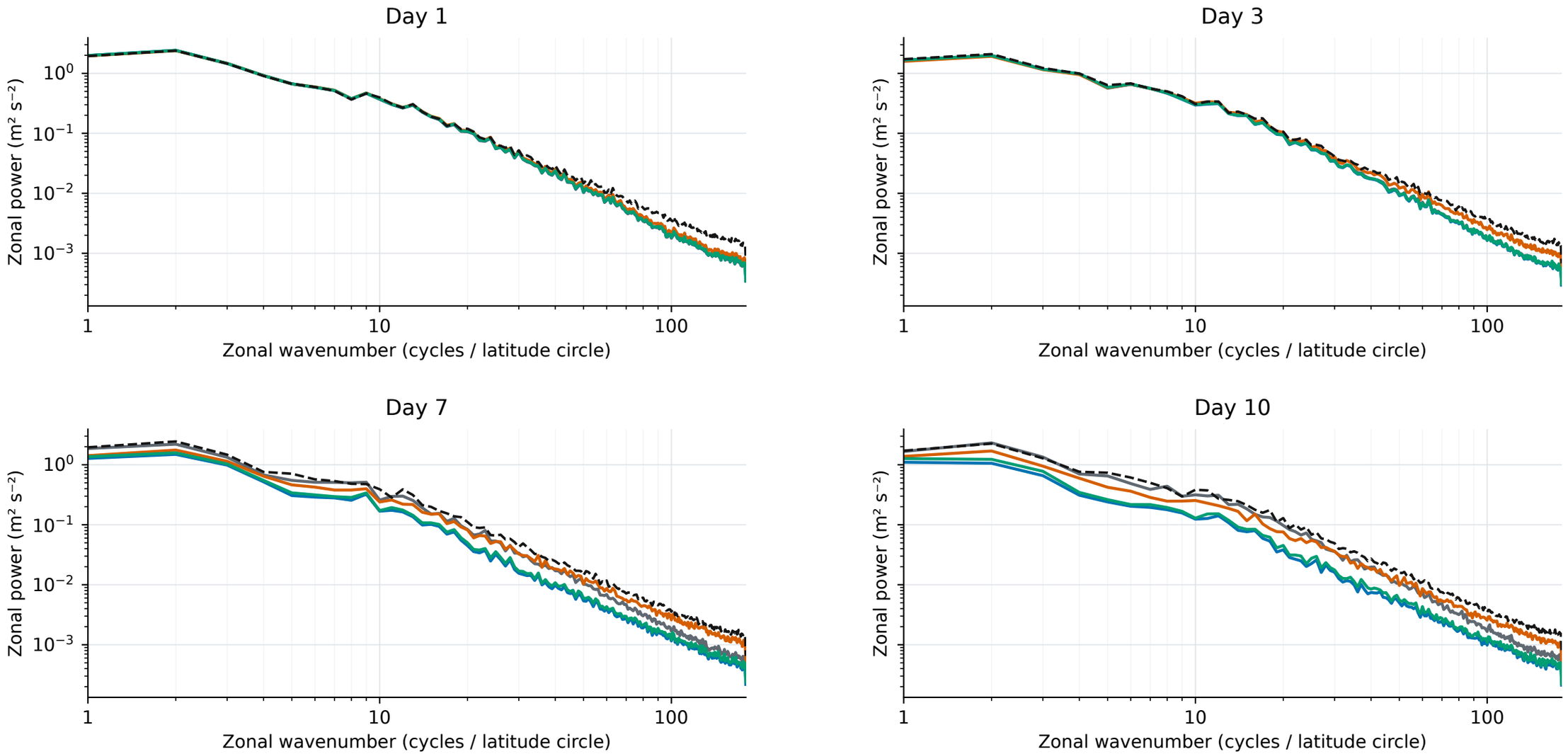
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m wind speed (WS10; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



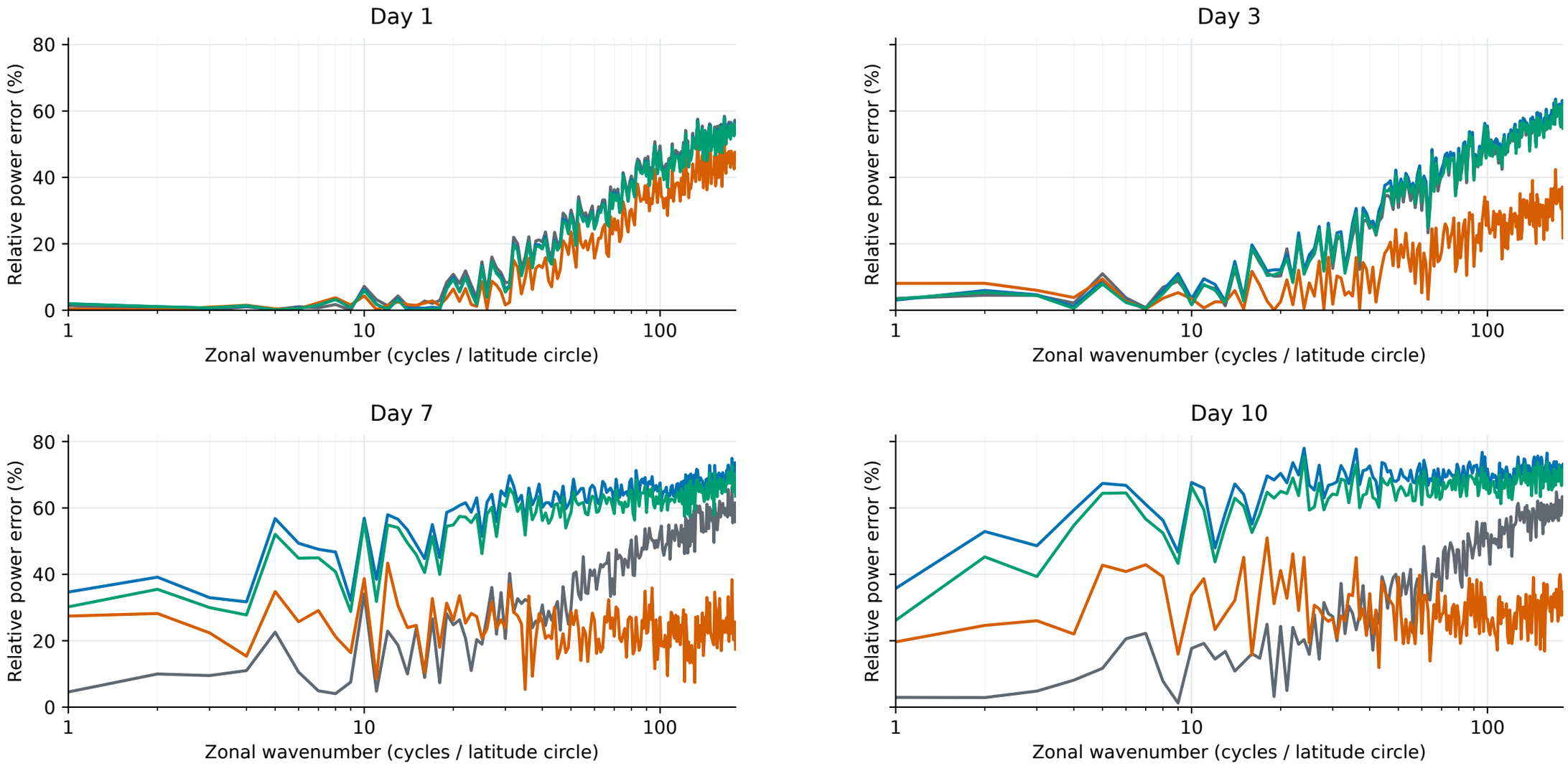
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m wind speed (WS10; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



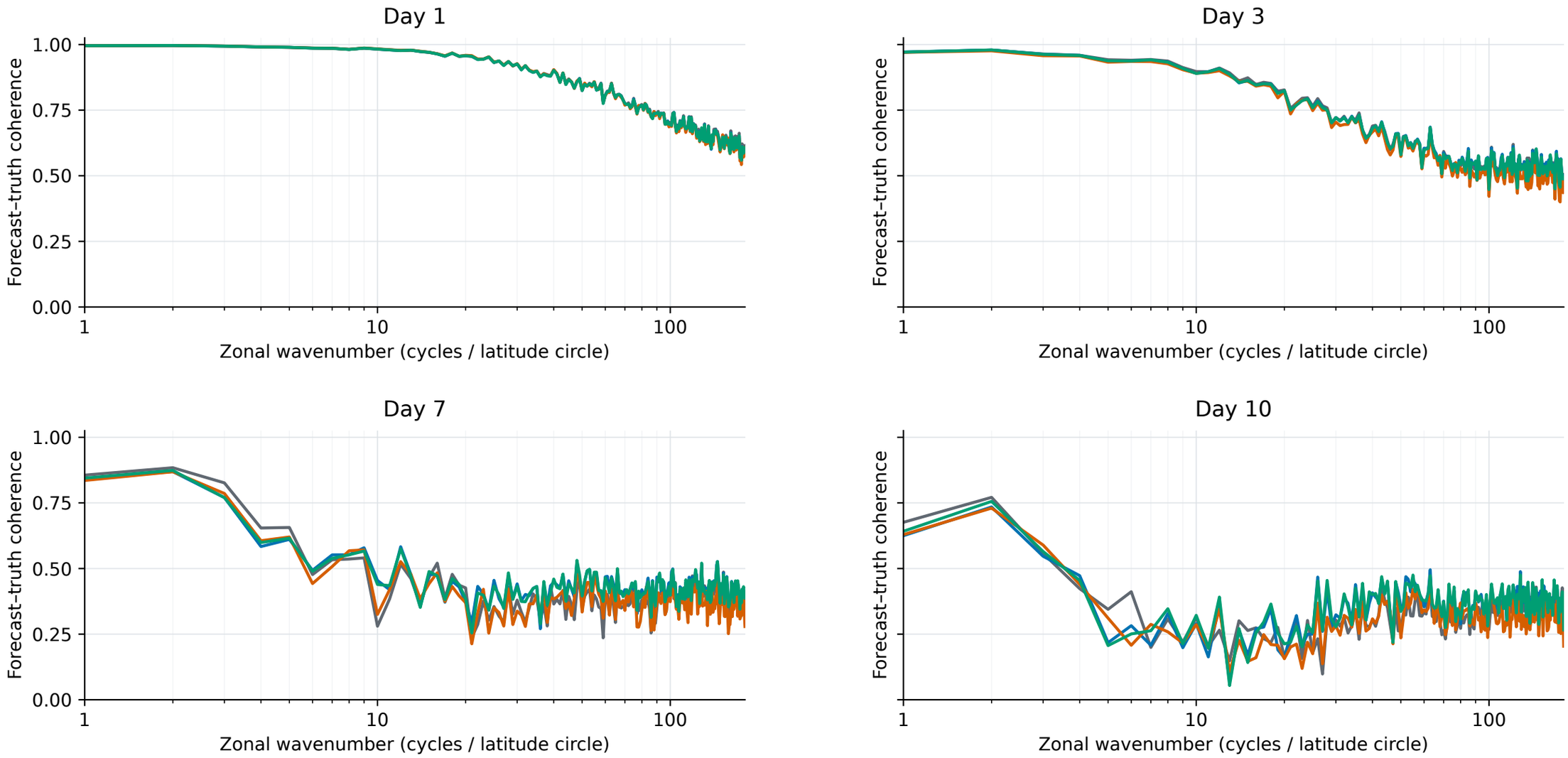
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m wind speed (WS10; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



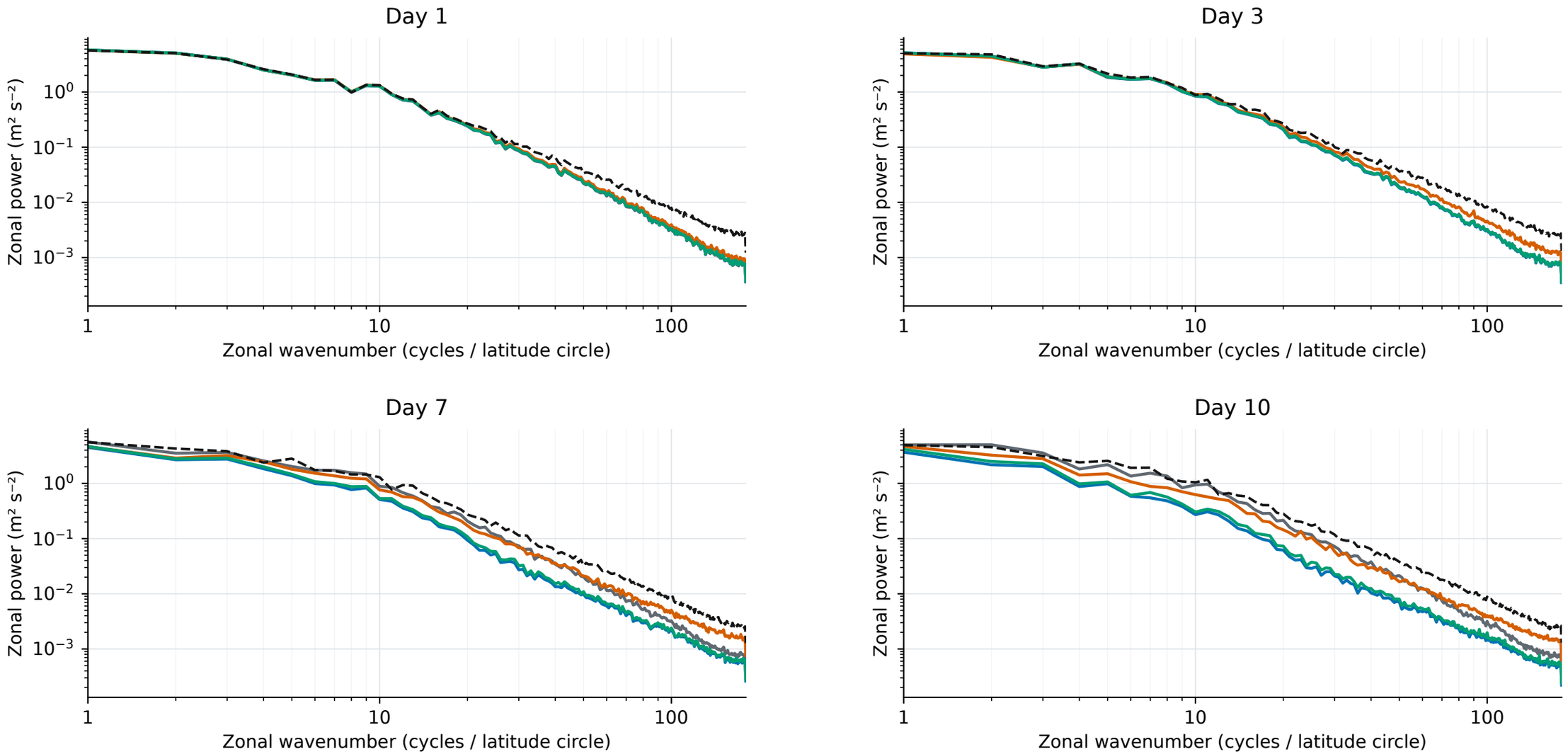
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Wind speed at 850 hPa (WS850; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



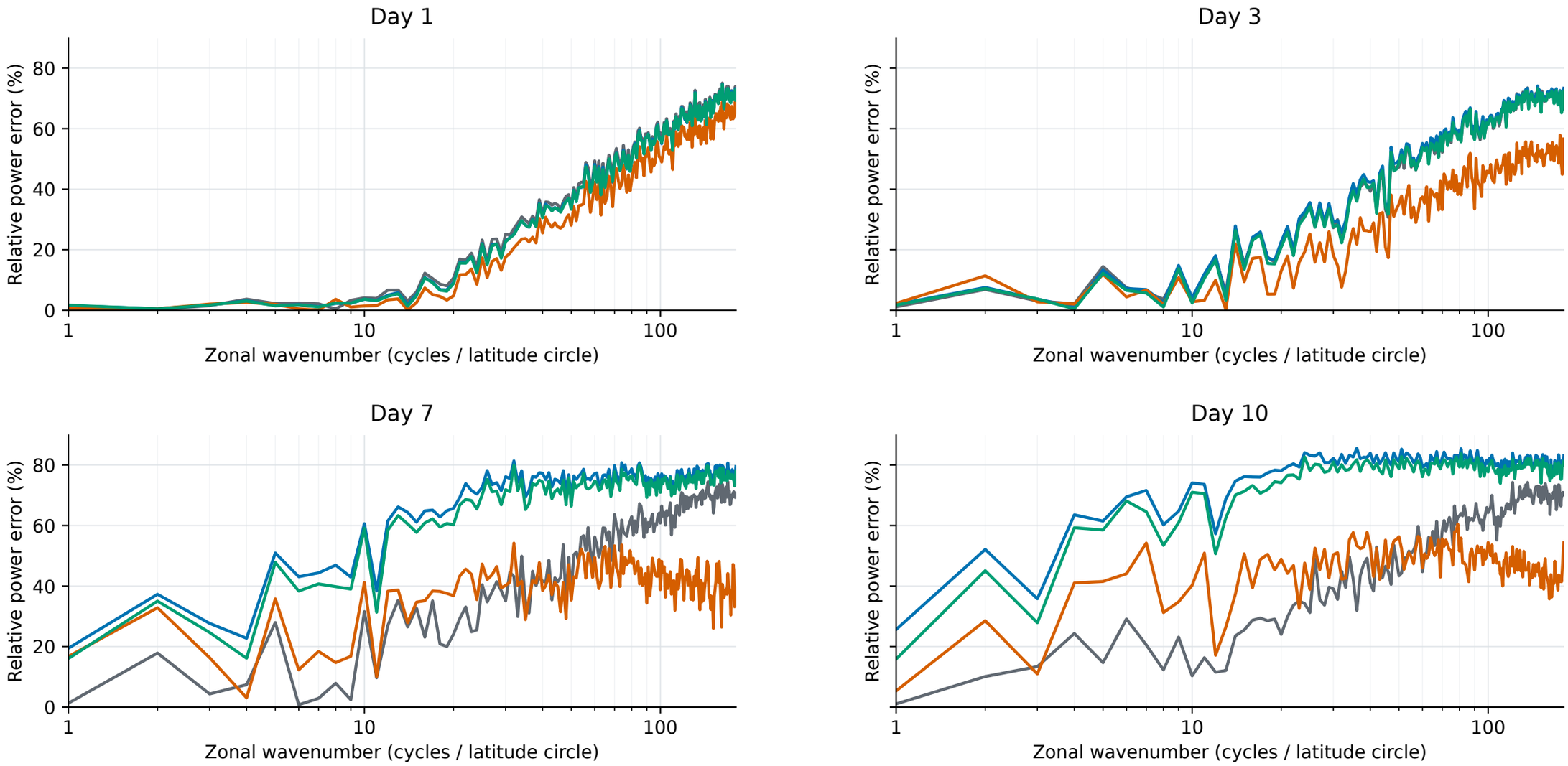
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Wind speed at 850 hPa (WS850; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



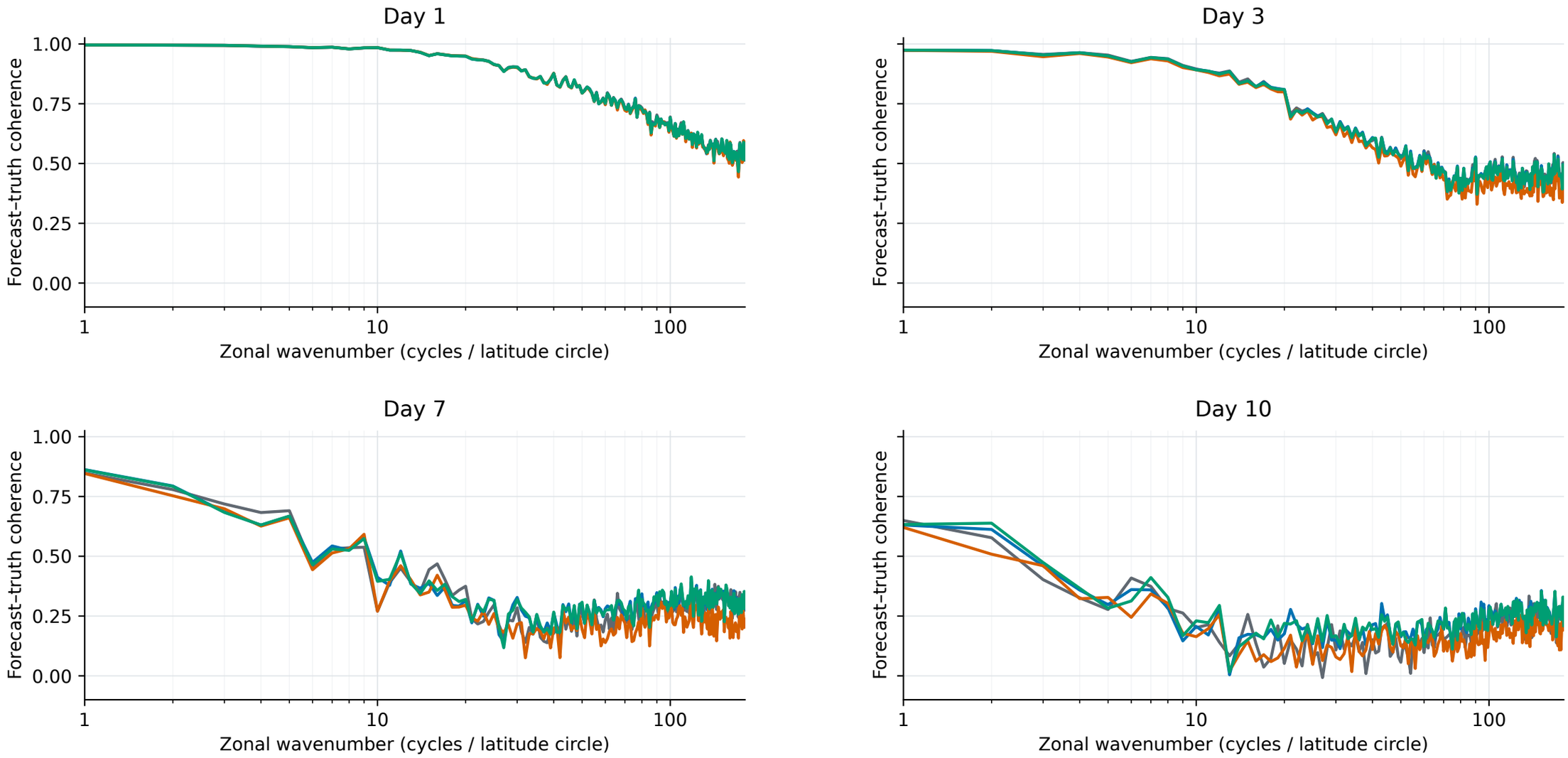
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Wind speed at 850 hPa (WS850; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Reference repeatability and interpretation limits

The figures contain exactly four forecast models

Year	Model	Exact GC rollout loss	vs displayed GC	vs own paired GC
2022	Original GC	10.814070	+0.000%	+0.000%
2022	MSE	9.130295	+15.570%	+15.570%
2022	AMSE λ=0.1	9.150518	+15.383%	+15.404%
2022	Full AMSE	10.818009	-0.036%	-0.017%
2023	Original GC	11.205208	+0.000%	+0.000%
2023	MSE	9.787777	+12.650%	+12.650%
2023	AMSE λ=0.1	9.779006	+12.728%	+13.055%
2023	Full AMSE	11.677263	-4.213%	-4.685%

Repeated original-GC forecasts exhibit numerical drift. All primary plots use one explicitly identified Original GC run, without averaging. The rightmost column retains each model's paired reference to show the practical effect of that choice. The complete audit also records the extra MSE control and per-lead baseline differences.

The spectral sample contains only eight previously inspected starts. The 2022 set contains 32 historical development starts. No confidence intervals, ensemble metrics or unseen test-set claims are made. Pressure-level, latitude-domain and weighting differences must be respected when comparing spectral and global-error charts.

Reproducibility and artifact guide

All source evaluations and checkpoints are preserved

Primary checkpoint policy: SWA of updates 500, 1000, 1500 and 2000. Inference uses 40 six-hour leads, cold initial recurrent state and full corrected feedback. Source manifests identify the original GraphCast checkpoint, data, initialization dates and metric settings.

The report_data.json and spectra.npz files contain the validated four-model inputs. CSV files provide all plotted spectral summaries, all 40 leads of global RMSE and every one of the 83 output channels. Source and checkpoint identities are recorded by SHA256. Standalone PNG and vector PDF files are provided for each figure.

Data and audit directory:

plots/analyze_models/data/resolution_eval/v24_amse_feature_report_20261004

Figure directory:

plots/analyze_models/images/resolution_eval/v24_amse_feature_report_20261004

Generators:

scripts/analyze_models/amse_feature_report_data.py

scripts/analyze_models/plot_v24_amse_feature_report.py

scripts/analyze_models/plot_v24_amse_feature_errors.py

scripts/analyze_models/report_v24_amse_features.py

Rebuild in that order after sourcing scripts/graphcast_env.sh. The process reads saved evaluations, validates them, generates plots and assembles this report. It does not submit training or evaluation jobs.